

CSE106 Discrete Mathematics

Credit Hours: $3 + 0 = 3$

**Prerequisite: CSE103 Structured Programming
Part II**

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Text book:

Kenneth H. Rosen, *Discrete Mathematics and Its Applications with Combinatorics and Graph Theory*, 7th Edition, Tata McGraw-Hill Publishing Company Limited, New Delhi, 2015.

Functions

Functions

Definition 1:

Let A and B be sets. A *function* f from A to B is an assignment of exactly one element of B to each element of A . We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A . If f is a function from A to B , we write $f: A \rightarrow B$.

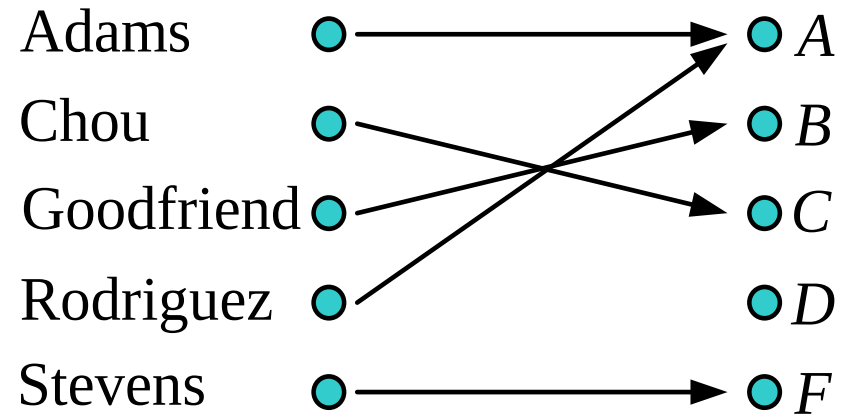


Figure: Assignment of Grades in a Discrete Mathematics Class.

Definition 2:

If f is a function from A to B , we say that A is the *domain* of f and B is the *codomain* of f .

If $f(a) = b$, we say that b is the *image* of a and a is a *pre-image* of b .

The *range* of f is the set of all images of elements of A .

Also, if f is a function from A to B , we say that f maps A to B .

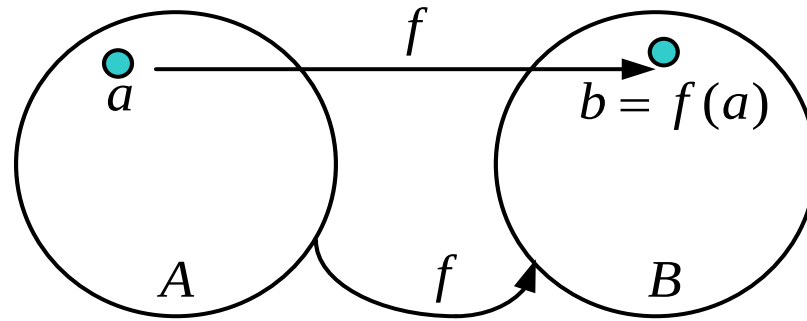
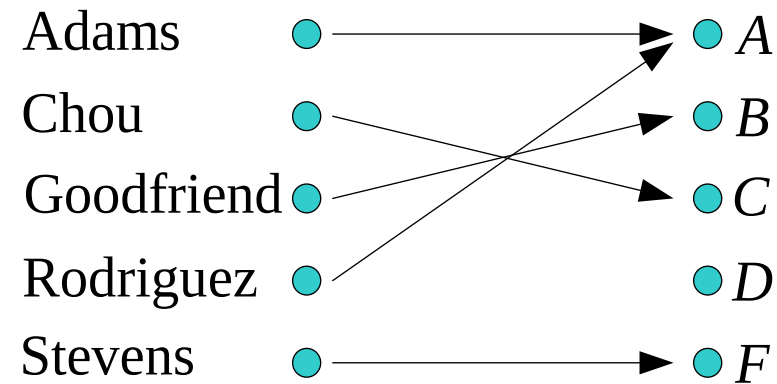


Figure: The function f maps A to B .

Example: What are the domain, codomain, and range of the function that assigns grades to students in discrete mathematics class described in the following figure?



Solution:

Let G be the function that assigns a grade to a student in discrete mathematics class.

$$G(\text{Adams}) = A.$$

The domain of G is the set $\{\text{Adams, Chou, Goodfriend, Rodriguez, Stevens}\}$

The codomain of G is the set $\{A, B, C, D, F\}$

The range of G is the set $\{A, B, C, F\}$.

Example: Let R be the relation consisting of ordered pairs (Abdul, 22), (Brenda, 24), (Carla, 21), (Desire, 22), (Eddie, 24), and (Felicia, 22), where each pair consists of a graduate student and the age of this student. What is the function that this relation determines?

Solution:

This relation defines the function f , where

$$f(\text{Abdul}) = 22$$

$$f(\text{Brenda}) = 24$$

$$f(\text{Carla}) = 21$$

$$f(\text{Desire}) = 22$$

$$f(\text{Eddie}) = 24$$

$$f(\text{Felicia}) = 22.$$

Here the domain is the set

{Abdul, Brenda, Carla, Desire, Eddie, Felicia}.

The codomain is the set of positive integers.

The range is the set {21, 22, 24}.

Example: Let $f: \mathbf{Z} \rightarrow \mathbf{Z}$ assign the square of an integer to an integer. Then, $f(x) = x^2$, where the domain of f is the set of all integers, the codomain of f is the set of all integers, and the range of f is the set of all integers that are perfect squares, namely, $\{0, 1, 4, 9, \dots\}$.

Definition 3: Let f_1 and f_2 be functions from A to $\mathbf{R}$. Then $f_1 + f_2$ and $f_1 f_2$ are also functions from A to $\mathbf{R}$ defined by

$$(f_1 + f_2)(x) = f_1(x) + f_2(x),$$
$$(f_1 f_2)(x) = f_1(x) f_2(x).$$

Example: Let f_1 and f_2 be functions from $\mathbf{R}$ to $\mathbf{R}$ such that $f_1(x) = x^2$ and $f_2(x) = x - x^2$. What are the functions $f_1 + f_2$ and $f_1 f_2$?

Solution:

$$(f_1 + f_2)(x) = f_1(x) + f_2(x) = x^2 + (x - x^2) = x$$
$$(f_1 f_2)(x) = f_1(x) f_2(x) = x^2 (x - x^2) = x^3 - x^4$$

One-to-One and Onto Functions

Definition 5: A function f is said to be *one-to-one*, or *injective*, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f .

Example: Determine whether the function f from $\{a, b, c, d\}$ to $\{1, 2, 3, 4, 5\}$ with $f(a) = 4$, $f(b) = 5$, $f(c) = 1$, and $f(d) = 3$ is one-to-one.

Solution:

The function f is one-to-one, since f takes on different values at the four elements of its domain.

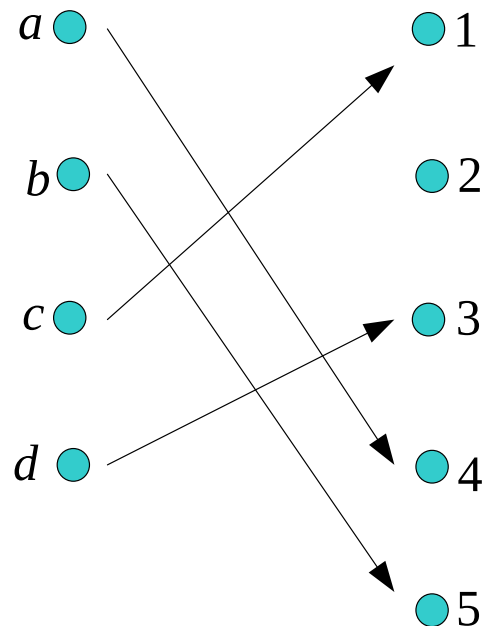


Figure 3 A one-to-one Function.

Example: Determine whether the function $f(x) = x^2$ from the set of integers to the set of integers is one-to-one.

Solution:

The function $f(x) = x^2$ is not one-to-one, since $f(1) = f(-1) = 1$, but $1 \neq -1$.

Example: Determine whether the function $f(x) = x + 1$ from the set of real numbers to itself is one-to-one.

Solution:

$$f(x) = x + 1$$

$$f(y) = y + 1$$

If $f(x) = f(y)$, then $x + 1 = y + 1$.

This implies that $x = y$.

Therefore, the function $f(x) = x + 1$ is one-to-one.

Definition 7: A function f from A to B is called *onto*, or *surjective*, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$.

Example: Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3\}$ defined by $f(a) = 3$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$. Is f an onto function?

Solution:

Since all three elements of the codomain are images of elements in the domain, f is onto.

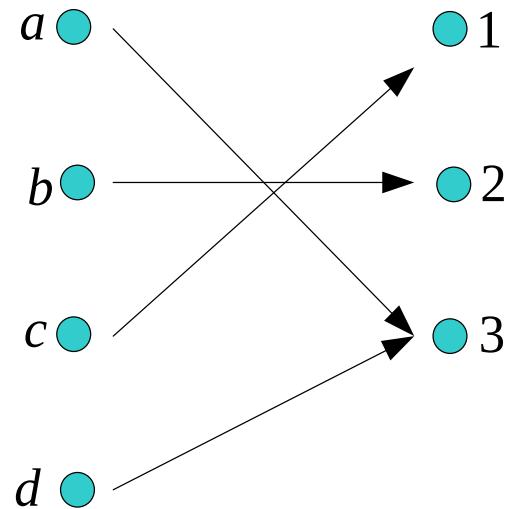


Figure: An Onto Function.

Example: Is the function $f(x) = x^2$ from the set of integers to the set of integers onto?

Solution:

The function f is not onto since there is no integer x with $x^2 = -1$, for example.

Example: Is the function $f(x) = x + 1$ from the set of integers to the set of integers onto?

Solution:

This function is onto, since for every integer y from the codomain, there is an integer x in the domain such that $f(x) = y = x + 1$.

Definition 8: The function f is a *one-to-one correspondence*, or a *bijective*, if it is both one-to-one and onto.

Example: Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3, 4\}$ with $f(a) = 4$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$. Is f a bijective function?

Solution:

$$f(a) = 4$$

$$f(b) = 2$$

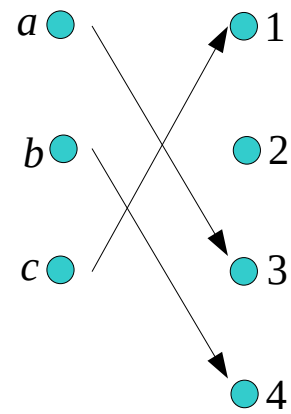
$$f(c) = 1$$

$$f(d) = 3$$

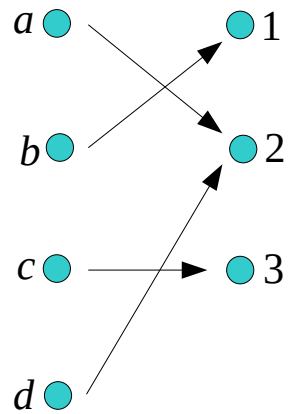
Each element of the domain maps to unique element of the codomain. So, the function f is one-to-one.

All elements of the codomain are mapped. So, the function f is onto.

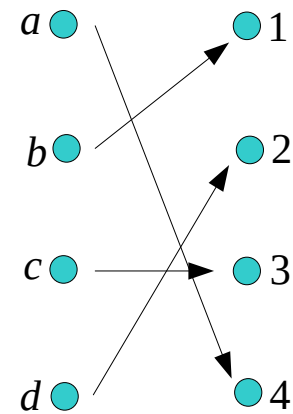
As the function f is both one-to-one and onto function, it is a bijective function.



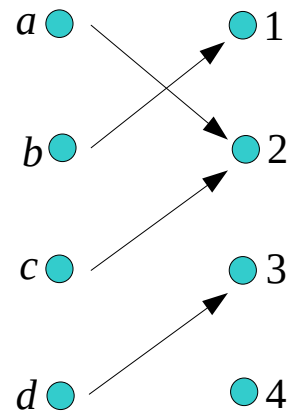
one - to - one,
not onto



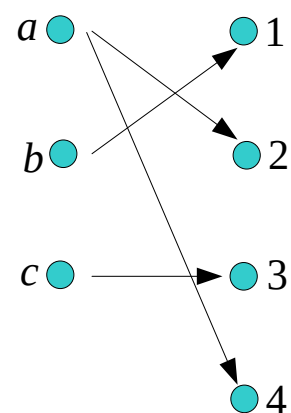
onto,
not one - to - one



one - to - one,
and onto



Neither one - to - one,
nor onto



Not a function

Figure: Examples of Different Types of Correspondance.

Inverse Functions and Compositions of Functions

Definition 9: Let f be a bijective function from the set A to the set B . The *inverse function* of f is the function that assigns to an element b belonging to B the unique element a in A such that $f(a) = b$. The inverse function of f is denoted by f^{-1} . Hence, $f^{-1}(b) = a$ when $f(a) = b$.

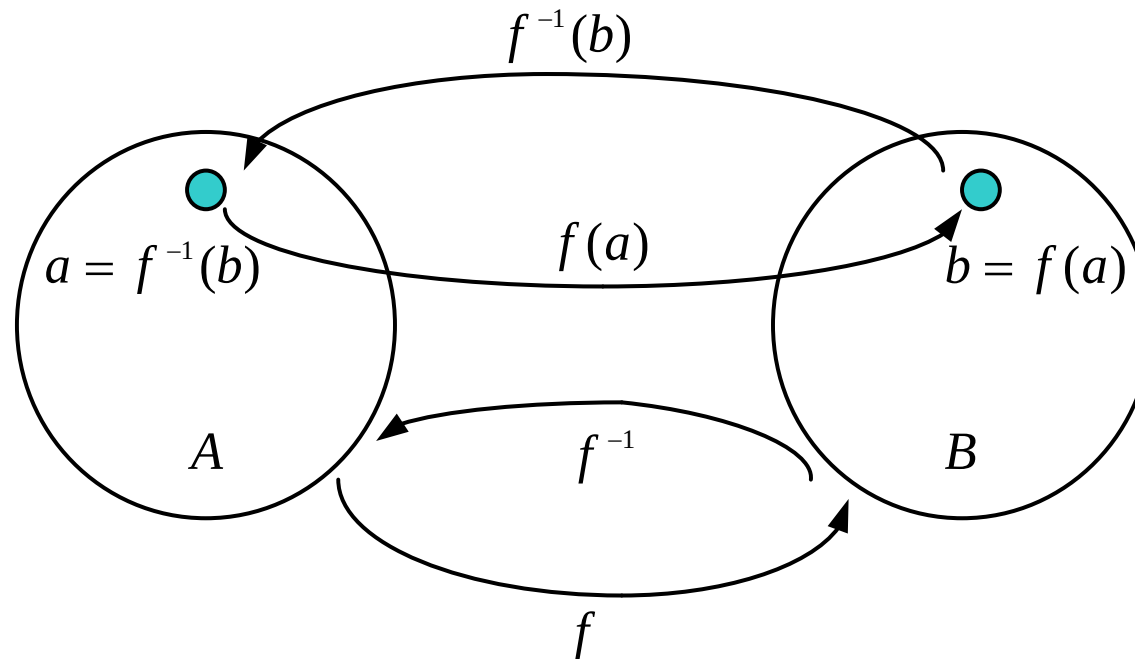


Figure: The Function f^{-1} is the Inverse of Function f .

- If a function f is not bijective, we cannot define an inverse function of f .
- A bijective function is called **invertible** because we can define an inverse of this function.
- A function is **not invertible** if it is not bijective function, because the inverse of such function does not exist.

Example: Let f be the function from $\{a, b, c\}$ to $\{1, 2, 3\}$ such that $f(a) = 2$, $f(b) = 3$, and $f(c) = 1$. Is f invertible, and if it is, what is its inverse?

Solution:

$$f(a) = 2$$

$$f(b) = 3$$

$$f(c) = 1$$

The function f is one-to-one and onto, so it is a bijective function and is invertible.

The inverse function f^{-1} is

$$f^{-1}(1) = c$$

$$f^{-1}(2) = a$$

$$f^{-1}(3) = b.$$

Example: Let $f: \mathbf{Z} \rightarrow \mathbf{Z}$ be such that $f(x) = x + 1$. Is f invertible, and if it is, what is its inverse?

Solution:

The function f is a one-to-one function.

The function f is an onto function.

So, the function f is a bijective function and is invertible.

Suppose that y is the image of x , so that $y = x + 1$.

Then $x = y - 1$. This means that $y - 1$ is the unique element of $\mathbf{Z}$ that is sent to y by f .

Consequently, $f^{-1}(y) = y - 1$.

Example: Let f be the function from $\mathbf{R}$ to $\mathbf{R}$ with $f(x) = x^2$. Is f invertible?

Solution:

Since $f(-2) = f(2) = 4$, f is not one-to-one. So, f is not a bijective function. Hence, f is not invertible.

Example: Let f be the function from the set of all nonnegative real numbers to the set of all nonnegative real numbers with $f(x) = x^2$. Show that f is invertible.

Solution:

If $f(x) = f(y)$, then $x^2 = y^2$, so $x^2 - y^2 = (x + y)(x - y) = 0$.

This means that $x + y = 0$ or $x - y = 0$, so $x = -y$ or $x = y$. Because both x and y are nonnegative, we must have $x = y$. so, $f(x) = x^2$ is one-to-one.

Furthermore, $f(x) = x^2$ is onto when the codomain is the set of all nonnegative real numbers, because each nonnegative real number has a square root. That is, if y is a nonnegative real number, there exists a nonnegative real number x such that $x = \sqrt{y}$, which means that $x^2 = y$.

Because the function $f(x) = x^2$ from the set of nonnegative real numbers to the set of nonnegative real numbers is one-to-one and onto, it is a bijective function and is invertible.

Definition 10: Let g be a function from the set A to the set B and let f be a function from the set B to the set C . The composition of the functions f and g , denoted by $f \circ g$, is defined by $(f \circ g)(a) = f(g(a))$.

To find $(f \circ g)(a)$ we first apply the function g to a to obtain $g(a)$ and then we apply the function f to the result of $g(a)$ to obtain $(f \circ g)(a) = f(g(a))$.

The composition $f \circ g$ cannot be defined unless the range of g is a subset of the domain of f .

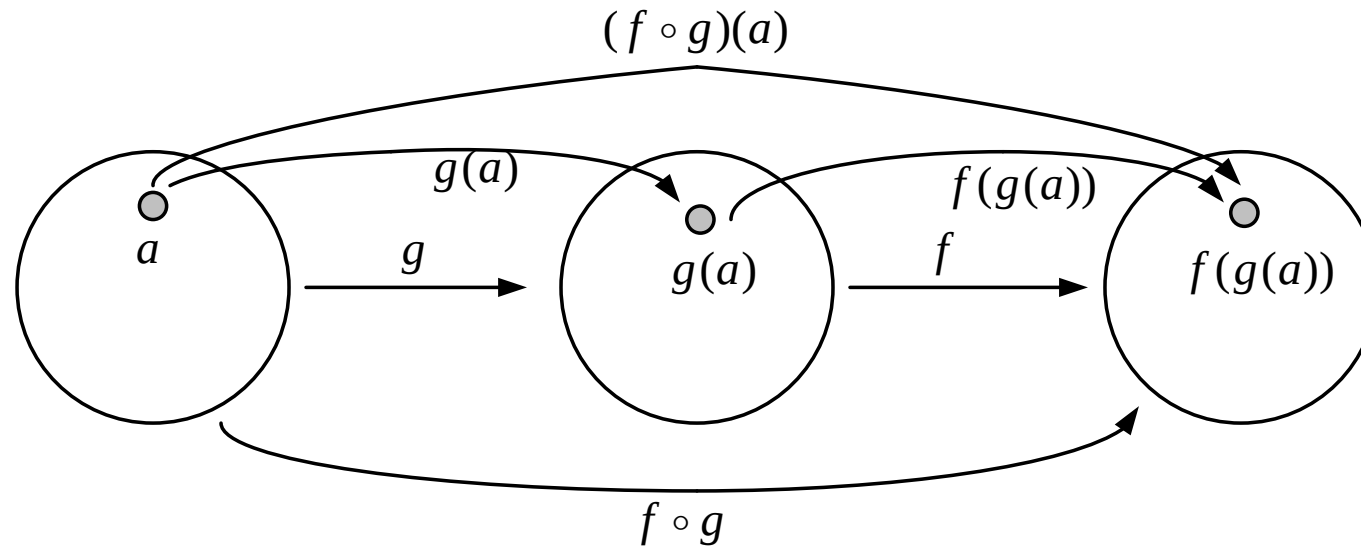


Figure: The Composition of the Functions f and g .

Example: Let g be the function from the set $\{a, b, c\}$ to itself such that $g(a) = b$, $g(b) = c$, and $g(c) = a$. Let f be the function from the set $\{a, b, c\}$ to the set $\{1, 2, 3\}$ such that $f(a) = 3$, $f(b) = 2$, and $f(c) = 1$. What is the composition of f and g , and what is the composition of g and f ?

Solution:

$$g(a) = b$$

$$g(b) = c$$

$$g(c) = a$$

$$f(a) = 3$$

$$f(b) = 2$$

$$f(c) = 1$$

The composition $f \circ g$ is defined by

$$(f \circ g)(a) = f(g(a)) = f(b) = 2,$$

$$(f \circ g)(b) = f(g(b)) = f(c) = 1,$$

$$(f \circ g)(c) = f(g(c)) = f(a) = 3.$$

$g \circ f$ is not defined, because the range of f is not a subset of the domain of g .

Example: Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g ? What is the composition of g and f ?

Solution:

$$(f \circ g)(x) = f(g(x)) = f(3x + 2) = 2(3x + 2) + 3 = 6x + 7$$

$$(g \circ f)(x) = g(f(x)) = g(2x + 3) = 3(2x + 3) + 2 = 6x + 11$$

Recursive Function

Recursively Defined Functions

We use two steps to define a function with the set of nonnegative integers as its domain:

BASIS STEP: Specify the value of the function at zero.

RECURSIVE STEP: Derive a rule for finding its value at an integer from its values at smaller integers.

Such a definition is called a **recursive** or **inductive** definition.

Example: Suppose that f is defined recursively by

$$f(0) = 3$$

$$f(n) = 2f(n-1) + 3 \text{ for } n = 1, 2, 3, \dots$$

Find $f(1)$, $f(2)$, $f(3)$, and $f(4)$.

Solution:

$$f(0) = 3$$

$$f(1) = 2f(0) + 3 = 2 \cdot 3 + 3 = 9$$

$$f(2) = 2f(1) + 3 = 2 \cdot 9 + 3 = 21$$

$$f(3) = 2f(2) + 3 = 2 \cdot 21 + 3 = 45$$

$$f(4) = 2f(3) + 3 = 2 \cdot 45 + 3 = 93$$

Example: Give a recursive definition of the factorial function $F(n) = n!$.

Solution:

$$F(0) = 1$$

$$F(n) = n \cdot F(n - 1) \text{ for } n = 1, 2, 3, \dots$$

$$F(5) = 5 \cdot F(4)$$

$$= 5 \cdot 4 \cdot F(3)$$

$$= 5 \cdot 4 \cdot 3 \cdot F(2)$$

$$= 5 \cdot 4 \cdot 3 \cdot 2 \cdot F(1)$$

$$= 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot F(0)$$

$$= 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 1$$

$$= 120$$

Example: Give a recursive definition of a^n , where a is a nonzero real number and n is a nonnegative integer.

Solution:

$$a^0 = 1$$

$$a^n = a \cdot a^{n-1} \text{ for } n = 1, 2, 3, \dots$$

Example: The *Fibonacci numbers*, $f_0, f_1, f_2, \dots$, are defined by the equations

$$f_0 = 0$$

$$f_1 = 1$$

$$f_n = f_{n-1} + f_{n-2} \text{ for } n = 2, 3, 4, \dots$$

Find the Fibonacci numbers f_2, f_3, f_4, f_5 , and f_6 .

Solution:

$$f_0 = 0$$

$$f_1 = 1$$

$$f_2 = f_1 + f_0 = 1 + 0 = 1$$

$$f_3 = f_2 + f_1 = 1 + 1 = 2$$

$$f_4 = f_3 + f_2 = 2 + 1 = 3$$

$$f_5 = f_4 + f_3 = 3 + 2 = 5$$

$$f_6 = f_5 + f_4 = 5 + 3 = 8$$

Relations

Relations and Their properties

Defintion 1: Let A and B be sets. A *binary relation from A to B* is a subset of $A \times B$.

A binary relation from A to B is a set R of ordered pairs where the first element of each ordered pair comes from A and the second element comes from B .

We use the notation aRb to denote that $(a, b) \in R$.

When (a, b) belongs to R , a is said to be **related** to b by R .

Example: Let $A = \{0, 1, 2\}$ and $B = \{a, b\}$.

Then $R = \{(0, a), (0, b), (1, a), (2, b)\}$ is a relation from A to B .

This means that $0Ra$.

Relations can be represented graphically, as shown in Figure 1, using arrows to represent ordered pairs. Another way to represent this relation is to use a table, which is also done in Figure 1.

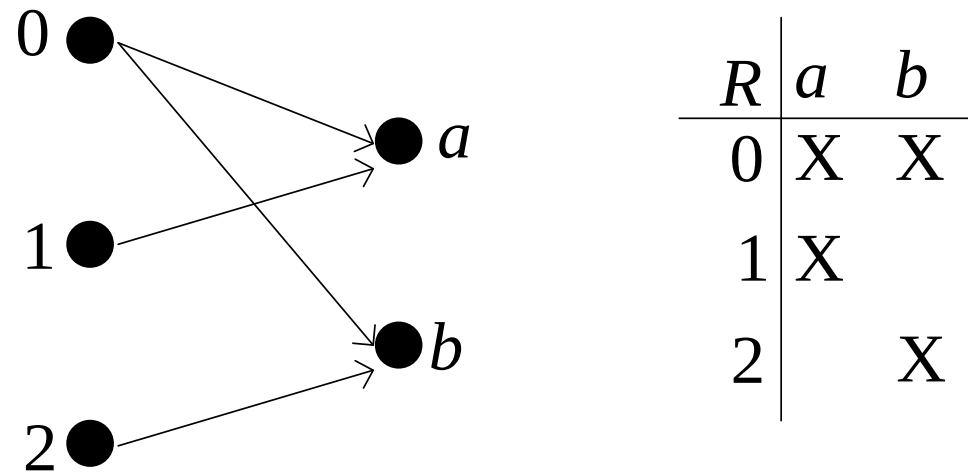


Figure: Displaying the Ordered Pairs in the Relation R from Example 3.

Functions as Relations

A function f from a set A to a set B assigns exactly one element of B to each element of A . The graph of f is the set of ordered pairs (a, b) such that $b = f(a)$. Since the graph of f is a subset of $A \times B$, it is a relation from A to B . Moreover, the graph of a function has the property that every element of A is the first element of exactly one ordered pair of the graph.

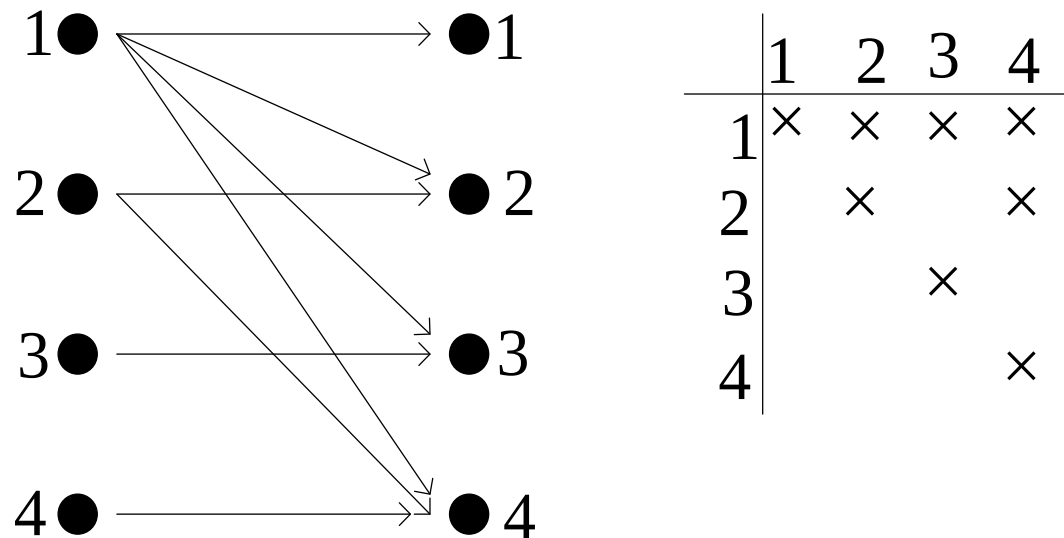
Relations on a Set

Definition 2: A relation on the set A is a relation from A to A .

Example: Let A be the set $\{1, 2, 3, 4\}$. Which ordered pairs are in the relation $R = \{(a, b) \mid a \text{ divides } b\}$?

Solution:

$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$



Example: Consider these relations on the set of integers:

$$R_1 = \{(a, b) \mid a \leq b\}$$

$$R_2 = \{(a, b) \mid a > b\}$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\}$$

$$R_4 = \{(a, b) \mid a = b\}$$

$$R_5 = \{(a, b) \mid a = b + 1\}$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}$$

Which of these relations contain each of the pairs (1, 1), (1, 2), (2, 1), (1, -1), and (2, 2)?

Solution:

(1, 1) is in R_1, R_3, R_4, R_6

(1, 2) is in R_1, R_6

(2, 1) is in R_2, R_5, R_6

(1, -1) is in R_2, R_3, R_6

(2, 2) is in R_1, R_3, R_4

Properties of Relations on sets

Definition 3: A relation R on a set A is called *reflexive* if $(a, a) \in R$ for every element $a \in A$.

The relation R on the set A is reflexive if $\forall a \in A ((a, a) \in R)$.

A relation on A is reflexive if every element of A is related to itself.

Example: Consider the following relations on $\{1, 2, 3, 4\}$:

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$$

$$R_2 = \{(1, 1), (1, 2), (2, 1)\}$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$$

$$R_6 = \{(3, 4)\}$$

Which of these relations are reflexive?

Solution:

The following relations are reflexive:

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$$

Example: Which of the following relations on integers are reflexive?

$$R_1 = \{(a, b) \mid a \leq b\}$$

$$R_2 = \{(a, b) \mid a > b\}$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\}$$

$$R_4 = \{(a, b) \mid a = b\}$$

$$R_5 = \{(a, b) \mid a = b + 1\}$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}$$

Solution:

The following relations are reflexive:

$$R_1 = \{(a, b) \mid a \leq b\} \text{ (since } a \leq a \text{ for every integer } a)$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\} \text{ (since } a = a \text{ for every integer } a)$$

$$R_4 = \{(a, b) \mid a = b\} \text{ (since } a = a \text{ for every integer } a)$$

Example: Is the “divides” relation on the set of positive integers reflexive?

Solution:

Since $a \mid a$ whenever a is a positive integer, the “divides” relation is reflexive.

Definition 4: A relation R on a set A is called *symmetric* if and only if $(b, a) \in R$ whenever $(a, b) \in R$, for all $a, b \in A$.

The relation R on the set A is symmetric if
 $\forall a \in A \forall b \in A ((a, b) \in R \leftrightarrow (b, a) \in R)$.

Example: Which of the following relations on $\{1, 2, 3, 4\}$ are symmetric?

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$$

$$R_2 = \{(1, 1), (1, 2), (2, 1)\}$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$$

$$R_6 = \{(3, 4)\}$$

Solution:

The following relations are symmetric, because in each case (b, a) belongs to the relation whenever (a, b) does:

$$R_2 = \{(1, 1), (1, 2), (2, 1)\}$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$$

Example: Which of the following relations on integers are symmetric?

$$R_1 = \{(a, b) \mid a \leq b\}$$

$$R_2 = \{(a, b) \mid a > b\}$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\}$$

$$R_4 = \{(a, b) \mid a = b\}$$

$$R_5 = \{(a, b) \mid a = b + 1\}$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}$$

Solution:

The following relations are symmetric:

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\} \text{ [If } a = b, \text{ then } b = a. \text{ If } a = -b \text{ then, } b = -a]$$

$$R_4 = \{(a, b) \mid a = b\} \text{ [If } a = b \text{ then, } b = a.]$$

$$R_6 = \{(a, b) \mid a + b \leq 3\} \text{ [If } a + b \leq 3 \text{ then, } b + a \leq 3]$$

The following relations are not symmetric:

$$R_1 = \{(a, b) \mid a \leq b\} \text{ [If } a \leq b \text{ then, } b \not\leq a]$$

$$R_2 = \{(a, b) \mid a > b\} \text{ [If } a > b \text{ then, } b \not> a]$$

$$R_5 = \{(a, b) \mid a = b + 1\} \text{ [If } a = b + 1 \text{ then, } b \neq a + 1]$$

Example: Is the “divides” relation on the set of positive integers symmetric?

Solution:

This relation is not symmetric, since $1 \mid 2$, but $2 \nmid 1$.

Definition 5: A relation R on a set A is called *transitive* if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$, for all $a, b, c \in A$.

The relation R on a set A is transitive if we have

$$\forall a \in A \forall b \in A \forall c \in A (((a, b) \in R \wedge (b, c) \in R) \rightarrow (a, c) \in R)$$

Example: Which of the following relations on $\{1, 2, 3, 4\}$ are transitive?

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$$

$$R_2 = \{(1, 1), (1, 2), (2, 1)\}$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$$

$$R_6 = \{(3, 4)\}$$

Solution:

The following relation is transitive:

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$$

$$[(3, 2) \text{ and } (2, 1) \rightarrow (3, 1)$$

$$(4, 2) \text{ and } (2, 1) \rightarrow (4, 1)$$

$$(4, 3) \text{ and } (3, 1) \rightarrow (4, 1)$$

$$(4, 3) \text{ and } (3, 2) \rightarrow (4, 2)]$$

Example (contd.):

The following relation is transitive:

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$$

$$[(1, 1) \text{ and } (1, 2) \rightarrow (1, 2)$$

$$(1, 1) \text{ and } (1, 3) \rightarrow (1, 3)$$

$$(1, 1) \text{ and } (1, 4) \rightarrow (1, 4)$$

$$(1, 2) \text{ and } (2, 2) \rightarrow (1, 2)$$

$$(1, 2) \text{ and } (2, 3) \rightarrow (1, 3)$$

$$(1, 2) \text{ and } (2, 4) \rightarrow (1, 4)$$

$$(1, 3) \text{ and } (3, 3) \rightarrow (1, 3)$$

$$(1, 3) \text{ and } (3, 4) \rightarrow (1, 4)$$

$$(1, 4) \text{ and } (4, 4) \rightarrow (1, 4)$$

$$(2, 2) \text{ and } (2, 3) \rightarrow (2, 3)$$

$$(2, 2) \text{ and } (2, 4) \rightarrow (2, 4)$$

$$(2, 3) \text{ and } (3, 3) \rightarrow (2, 3)$$

$$(2, 3) \text{ and } (3, 4) \rightarrow (2, 4)$$

$$(2, 4) \text{ and } (4, 4) \rightarrow (2, 4)$$

$$(3, 3) \text{ and } (3, 4) \rightarrow (3, 4)$$

$$(3, 4) \text{ and } (4, 4) \rightarrow (3, 4)]$$

Example (contd.):

The following relation is transitive:

$$R_6 = \{(3, 4)\}$$

[There is no pair like $(4, c)$ which would imply the existence of the pair $(3, c)$]

The following relations are not transitive:

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$$

[There are pairs $(3, 4)$ and $(4, 1)$ but not $(3, 1)$]

$$R_2 = \{(1, 1), (1, 2), (2, 1)\}$$

[There are pairs $(2, 1)$ and $(1, 2)$ but not $(2, 2)$]

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$$

[There are pairs $(4, 1)$ and $(1, 2)$ but not $(4, 2)$]

Example: Which of the following relations on integers are transitive?

$$R_1 = \{(a, b) \mid a \leq b\}$$

$$R_2 = \{(a, b) \mid a > b\}$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\}$$

$$R_4 = \{(a, b) \mid a = b\}$$

$$R_5 = \{(a, b) \mid a = b + 1\}$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}$$

Solution:

The following relations are transitive:

$$R_1 = \{(a, b) \mid a \leq b\} [(a \leq b) \text{ and } (b \leq c) \rightarrow (a \leq c)]$$

$$R_2 = \{(a, b) \mid a > b\} [(a > b) \text{ and } (b > c) \rightarrow (a > c)]$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\} [(a = \pm b) \text{ and } (b = \pm c) \rightarrow (a = \pm c)]$$

$$R_4 = \{(a, b) \mid a = b\} [(a = b) \text{ and } (b = c) \rightarrow (a = c)]$$

The following relations are not transitive:

$$R_5 = \{(a, b) \mid a = b + 1\} [(2, 1), (1, 0) \in R_5 \text{ but } (2, 0) \notin R_5]$$

$$R_6 = \{(a, b) \mid a + b \leq 3\} [(2, 1), (1, 2) \in R_6 \text{ but } (2, 2) \notin R_6]$$

Example: Is the “divides” relation on the set of positive integers transitive?

Solution:

If a divides b , then $b = ak$, where k is a positive integer

If b divides c , then $c = bl$, where l is a positive integer

Then $c = bl = akl = a(kl)$, where kl is a positive integer. Therefore, a divides c

Hence, the “divides” relation is transitive.

Counting

The Basics of Counting

Basic Counting principles

THE PRODUCT RULE

Suppose that a procedure can be broken down into a sequence of two tasks. If there are n_1 ways to do the first task and n_2 ways to do the second task after the first task has been done, then there are $n_1 n_2$ ways to do the procedure.

Example: A new company with just two employees, Sahin and Parvin, rents a floor of a building with 12 offices. How many ways are there to assign different offices to these two employees?

Solution:

An office can be assigned to Sahin in 12 ways.

After assigning an office to Sahin, an office can be assigned to Parvin in 11 different ways.

Therefore, by the product rule, there are $12 \cdot 11 = 132$ ways to assign offices to these two employees.

Example: The chairs of an auditorium are to be labeled with a letter and a positive integer not exceeding 100. What is the largest number of chairs that can be labeled differently?

Solution:

A letter can be assigned in 26 different ways

An integer can be assigned in 100 different ways

The different ways that a chair can be labeled is $26 \cdot 100 = 2600$

Therefore, the largest number of chairs that can be labeled differently is 2600.

Example: How many different bit strings are there of length seven?

Solution:

Each of the seven bits can be chosen in two ways

Therefore, there are a total of $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^7 = 128$ different bit strings of length seven.

Example: What is the value of k after the following code has been executed?

```
 $k := 0$   
for  $i_1 := 1$  to  $n_1$   
    for  $i_2 := 1$  to  $n_2$   
        .  
        .  
        .  
        for  $i_m := 1$  to  $n_m$   
             $k := k + 1$ 
```

Solution:

The initial value of k is zero

Each time the nested loop is traversed, 1 is added to k

The nested loop is traversed $n_1 n_2 \dots n_m$ times

Therefore, the final value of k is $n_1 n_2 \dots n_m$

THE SUM RULE

If a first task can be done in n_1 ways and second task in n_2 ways, and if these tasks cannot be done at the same time, then there are $n_1 + n_2$ ways to do one of these tasks.

Example: Suppose that either a member of the mathematics faculty or a student who is mathematics major is chosen as a representative to a university committee. How many different choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors?

Solution:

Choosing a member of the mathematics faculty can be done in 37 ways.

Choosing mathematics major can be done in 83 ways.

There are $37 + 83 = 120$ possible ways to pick this representative.

Example: A student can choose a computer project from one of three lists. The three lists contain 23, 15, and 19 possible projects, respectively. How many possible projects are there to choose from?

Solution:

The student can choose a project from the first list in 23 ways, from the second list in 15 ways, and from the third list in 19 ways.

Hence, there are $23 + 15 + 19 = 57$ projects to choose from.

Example: What is the value of k after the following code has been executed?

```
 $k := 0$   
for  $i_1 := 1$  to  $n_1$   
     $k := k + 1$   
for  $i_2 := 1$  to  $n_2$   
     $k := k + 1$   
    .  
    .  
    .  
for  $i_m := 1$  to  $n_m$   
     $k := k + 1$ 
```

Solution:

The initial value of k is zero.

This block of code is made up of m different loops. Each time a loop is traversed, 1 is added to k .

The final value of k is $n_1 + n_2 + \dots + n_m$.

More Complex Counting problems

Example: In a version of the computer language BASIC, the name of a variable is a string of one or two alphanumeric characters, where uppercase and lowercase letters are not distinguished. Moreover, a variable name must begin with a letter and must be different from the five strings of two characters that are reserved for programming use. How many different variables names are there in this version of BASIC?

Solution:

Let V be the number of different variable names in this version of BASIC.

Let V_1 be the number of these that are one character long.

$$V_1 = 26$$

Let V_2 be the number of these that are two characters long

$$V_2 = 26 \cdot 36 - 5 = 931$$

$$V = V_1 + V_2 = 26 + 931 = 957$$

Example: Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

Solution:

Let P be the total number of possible passwords, and P_6 , P_7 , and P_8 denote the number of possible passwords of length 6, 7, and 8, respectively.

The number of strings of six characters is 36^6

The number of strings with no digits is 26^6

$$\therefore P_6 = 36^6 - 26^6 = 2,176,782,336 - 308,915,776 = 1,867,866,560$$

Similarly,

$$P_7 = 36^7 - 26^7 = 78,367,164,096 - 8,031,810,176 = 70,332,353,920$$

$$P_8 = 36^8 - 26^8 = 2,821,109,907,456 - 208,827,064,576 = 2,612,282,842,880$$

$$P = P_6 + P_7 + P_8 = 1,867,866,560 + 70,332,353,920 + 2,612,282,842,880 = 2,684,483,063,360$$

The Inclusion-Exclusion Principle

When a part of two tasks can be done at the same time, we cannot use the sum rule to count the number of ways to do one of the two tasks. Adding the number of ways to do each task leads to an overcount, since the ways to do the part in both tasks are counted twice. To correctly count the number of ways to do one of the two tasks, we add the number of ways to do each of the two tasks and then subtract the number of ways to do the part of both tasks. This technique is called the **principle of inclusion-exclusion**.

Example: How many bit strings of length eight start with a 1 bit or end with the two bits 00?

Solution:

Constructing a bit string of length eight beginning with a 1 bit can be done in $2^7 = 128$ ways.

Constructing a bit string of length eight ending with the two bits 00 can be done in $2^6 = 64$ ways.

Constructing a bit string of length eight that begins with a 1 and ends with 00 can be done in $2^5 = 32$ ways.

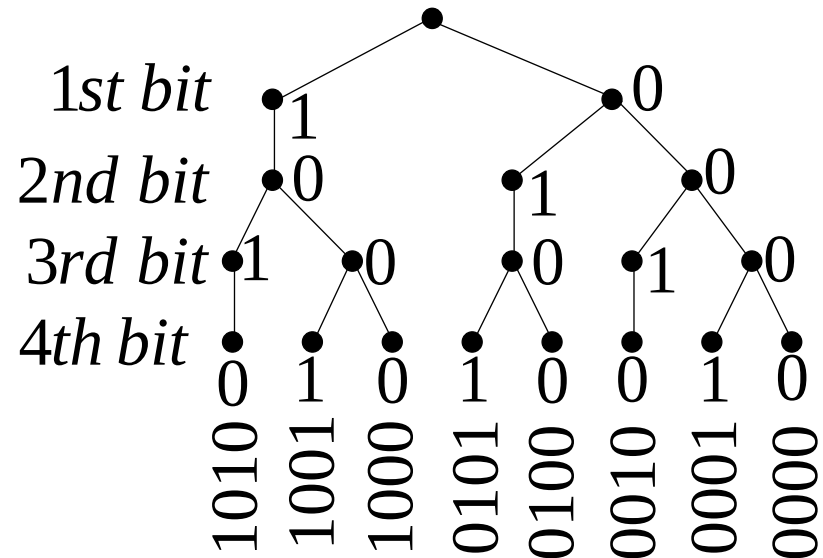
Consequently, the number of bit strings of length eight that begin with a 1 or end with a 00 equals $128 + 64 - 32 = 160$.

Tree Diagrams

Counting problems can be solved using **tree diagrams**. To use trees in counting, we use a branch to represent each possible choice. We represent the possible outcomes by the leaves.

Example: How many bit string of length four do not have two consecutive 1s?

Solution:



The tree diagram displays all bit strings of length four without two consecutive 1s. We see that there are eight bit strings of length four without two consecutive 1s.

The Pigeonhole Principle

Theorem 1: THE PIGEONHOLE PRINCIPLE

If k is a positive integer and $k + 1$ or more objects are placed into k boxes, then there is **at least one box** containing two or more (**at least two**) of the objects.

Proof.

Suppose that none of the k boxes contains more than one object. Then the total number of objects would be at most k . This is a contradiction, since there are at least $k + 1$ objects.

Example: Among any group of 367 people, there must be at least two with the same birthday, because there are only 366 possible birthdays.

Example: In any group of 27 English words, there must be at least two that begin with the same letter, since there are 26 letters in the English alphabet.

Example: How many students must be in a class to guarantee that at least two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 100?

Solution:

There are 101 possible scores on the final. The pigeonhole principle shows that among any $101 + 1 = 102$ students there must be at least 2 students with the same score.

The Generalized Pigeonhole principle

Theorem 2: THE GENERALIZED PIGEONHOLE PRINCIPLE

If N objects are placed into k boxes, then there is at least one box containing at least $r = \lceil N/k \rceil$ objects.

Proof:

Suppose that none of the boxes contain more than $\lceil N/k \rceil - 1$ objects. Then, the total number of objects is at most

$$k \left(\left\lceil \frac{N}{k} \right\rceil - 1 \right) < k \left(\left(\frac{N}{k} + 1 \right) - 1 \right) = N$$

This is a contradiction, since there are a total of N objects.

A common type of problem asks for the minimum number of objects N so that at least r of these objects must be in one of k boxes when these objects are distributed among the boxes.

When we have N objects, the generalized pigeonhole principle tells us that there must be at least r objects in one of the boxes as long as

$$\lceil N/k \rceil \geq r$$

$$N/k + 1 > r$$

$$N/k > (r - 1)$$

$$N > k(r - 1)$$

$N = k(r - 1) + 1$ is the smallest integer satisfying the condition

Example: Among 100 people there are at least $\lceil 100/12 \rceil = 9$ who were born in the same month.

Example: What is the minimum number of students required in a discrete mathematics class to be sure that at least six will receive the same grade, if there are five possible grades, A, B, C, D, and F?

Solution:

Let

N = minimum number of students to satisfy the condition = ?

k = total number of grades = 5

r = minimum number of students receiving the same grade = 6

$$N = k(r - 1) + 1 = 5(6 - 1) + 1 = 26$$

Algorithms, the Integers

Algorithms

Definition 1: An *algorithm* is a finite set of precise instructions for performing a computation or for solving a problem.

Example: Describe an algorithm using **pseudocode** for finding the maximum (largest) value in a finite sequence of integers.

ALGORITHM 1: Finding the Maximum Element in a Finite Sequence.

procedure *max* ($a_1, a_2, \dots, a_n$: integers)

max := a_1

for $i := 2$ **to** n

if $max < a_i$ **then** $max := a_i$

{*max* is the largest element}

Searching Algorithms

Searching problem: Locate an element x in a list of distinct elements $a_1, a_2, \dots, a_n$, or determine that it is not in the list. The solution to this search problem is the location of the term in the list that equals x (that is, i is the solution if $x = a_i$) and is 0 if x is not in the list.

THE LINEAR SEARCH

ALGORITHM 2: The Linear Search Algorithm.

procedure *linear search* (x : integer, $a_1, a_2, \dots, a_n$: distinct integers)

$i := 1$

while ($i \leq n$ and $x \neq a_i$)

$i := i + 1$

if $i \leq n$ **then** $location := i$

else $location := 0$

{*location* is the subscript of the term that equals x , or is 0 if x is not found}

THE BINARY SEARCH

The **binary search algorithm** can be used when the list has terms occurring in order of increasing value.

Example: Search for 19 in the list

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Index
1	2	3	5	6	7	8	10	12	13	15	16	18	19	20	22	Data

$$m = \lfloor (1 + 16)/2 \rfloor = 8 \text{ (middle point)}$$

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	2	3	5	6	7	8	10	12	13	15	16	18	19	20	22

Since $19 > 10$, solution is in the second portion

9	10	11	12	13	14	15	16
12	13	15	16	18	19	20	22

$$m = \lfloor (9 + 16)/2 \rfloor = 12$$

9	10	11	12	13	14	15	16
12	13	15	16	18	19	20	22

Since $19 > 16$, solution is in the second portion

Example (contd.): Search for 19 in the list

13 14 15 16

18	19	20	22
----	----	----	----

$$m = \lfloor (13 + 16)/2 \rfloor = 14$$

13 14 15 16

18	19
----	----

20	22
----	----

Since 19 not > 19, solution is in the first portion

13 14

18	19
----	----

$$m = \lfloor (13 + 14)/2 \rfloor = 13$$

13 14

18

19

Since 19 > 18, solution is in the second portion

14

19

19 is the 14th term of the list. Return value is 14.

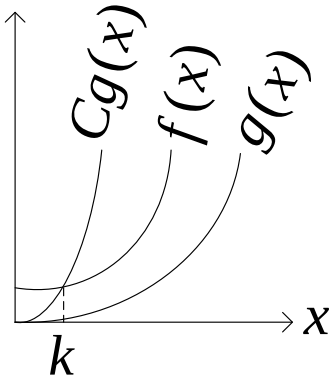
ALGORITHM 3: The Binary Search Algorithm.

```
procedure binary search ( $x$ : integer,  $a_1, a_2, \dots, a_n$ : increasing integers)
 $i := 1$       { $i$  is left endpoint of search interval}
 $j := n$       { $j$  is right endpoint of search interval}
while  $i < j$ 
begin
     $m := \lfloor (i + j)/2 \rfloor$       {  $m$  is middle point of search interval}
    if  $x > a_m$  then  $i := m + 1$ 
    else  $j := m$ 
end
if  $x = a_i$  then  $location := i$ 
else  $location := 0$ 
{ $location$  is the subscript of the term that equals  $x$ , or is 0 if  $x$  is not found}
```

The Growth of Functions

Big-O Notation

Definition 1: Let f and g be functions from the set of non-negative integers or the set of non-negative real numbers to the set of non-negative real numbers. We say that $f(x) = O(g(x))$ if there are constants C and k such that $f(x) \leq C g(x)$, whenever $x \geq k$.



The constants C and k in the definition of big- O notation are called **witness** to the relationship $f(x) = O(g(x))$.

To establish that $f(x) = O(g(x))$ we need only one pair of witnesses to this relationship.

Example: Show that $f(x) = x^2 + 2x + 1 = O(x^2)$.

Solution:

$$\begin{aligned} f(x) = x^2 + 2x + 1 &\leq x^2 + 2x^2 + x^2 && [x \leq x^2, 1 \leq x^2 \text{ for } x \geq 1] \\ &= 4x^2 \end{aligned}$$

When $k = 1$, $C = 4$, and $g(x) = x^2$, $f(x) = x^2 + 2x + 1 \leq 4x^2$.

$\therefore f(x) = x^2 + 2x + 1 = O(x^2)$.

When big- O notation is used, the function g in the relationship $f(x) = O(g(x))$ is chosen to be **as small as possible** (sometimes from a set of reference functions)

Some Important Big- O Results

Theorem 1: Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, where $a_0, a_1, \dots, a_{n-1}, a_n$ are real numbers with $a_n \neq 0$. Then $f(x) = O(x^n)$.

Proof: Omitted

Example: How can big- O notation be used to estimate the sum of the first n positive integers?

Solution:

$$1 + 2 + \dots + n \leq n + n + \dots + n = n^2$$

For $k = 1$ and $C = 1$

$$1 + 2 + \dots + n = O(n^2)$$

Example: Give big- O estimate for the factorial function and the logarithm of the factorial function, where the factorial function $f(n) = n!$ is defined by $n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n$, whenever n is a positive integer, and $1! = 1$.

Solution:

$$n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n \leq n \cdot n \cdot n \cdot \dots \cdot n = n^n$$

For $k = 1$ and $C = 1$

$$n! = O(n^n)$$

$$n! \leq n^n$$

$$\text{or, } \log n! \leq \log n^n = n \log n$$

For $k = 1$ and $C = 1$

$$\log n! = O(n \log n)$$

Big-O notation is used to estimate the number of operations needed to solve a problem using a specified procedure or algorithm. The functions used in these estimates often include the following:

$1, \log n, n, n \log n, n^2, 2^n, n!$

Each function in the list is smaller than the succeeding function.

Example:

Let $f(x) = c$, where c is a positive constant

$f(x) = c \leq c \cdot 1$, where $k = 0$, $C = c$, and $g(x) = 1$

$\therefore f(x) = c = O(1)$

Big-O of any constant is $O(1)$.

The Growth of Combination of Functions

Theorem 2: Suppose that $f_1(x) = O(g_1(x))$ and $f_2(x) = O(g_2(x))$. Then $(f_1 + f_2)(x) = O(\max(g_1(x), g_2(x)))$

Proof: Omitted

Theorem 3: Suppose that $f_1(x) = O(g_1(x))$ and $f_2(x) = O(g_2(x))$. Then $(f_1 f_2)(x) = O(g_1(x)g_2(x))$.

Proof: Omitted

Example: Give a big- O estimate for $f(n) = 3n\log(n!) + (n^2 + 3)\log n$, where n is a positive integer.

Solution:

$$3n = O(n)$$

$$\log(n!) = O(n\log n)$$

$$\therefore 3n\log(n!) = O(n (n\log n)) = O(n^2\log n)$$

$$(n^2 + 3) = O(n^2)$$

$$\log n = O(\log n)$$

$$\therefore (n^2 + 3)\log n = O(n^2 (\log n)) = O(n^2\log n)$$

$$\begin{aligned}\therefore f(n) &= 3n\log(n!) + (n^2 + 3)\log n \\ &= O(\max(n^2\log n, n^2\log n)) = O(n^2\log n)\end{aligned}$$

Example: Give big- O estimate for $f(x) = (x + 1)\log(x^2 + 1) + 3x^2$

Solution:

$$(x + 1) = O(x)$$

$$\begin{aligned}\log(x^2 + 1) &\leq \log(x^2 + x^2) \text{ [when } x \geq 1\text{]} \\ &= \log(2x^2) \\ &= \log 2 + \log x^2 \\ &= \log 2 + 2\log x \\ &\leq \log x + 2\log x \text{ [when } x \geq 2\text{]} \\ &= 3\log x\end{aligned}$$

$$\therefore \log(x^2 + 1) = O(\log x) \text{ [} k = 2 \text{ and } C = 3\text{]}$$

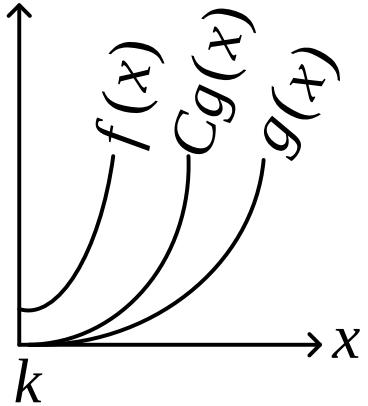
$$\therefore (x + 1)\log(x^2 + 1) = O(x\log x)$$

$$3x^2 = O(x^2)$$

$$\therefore f(x) = (x + 1)\log(x^2 + 1) + 3x^2 = O(\max(x\log x, x^2)) = O(x^2)$$

Big-Omega and Big-Theta Notation

Definition 2: Let f and g be functions from the set of non-negative integers or the set of non-negative real numbers to the set of non-negative real numbers. We say that $f(x) = \Omega(g(x))$ if there are constants C and k such that $f(x) \geq C g(x)$, where $x \geq k$.



Example:

$$f(x) = 8x^3 + 5x^2 + 7 \geq 8x^3 \text{ [when } x \geq 0]$$

$$\therefore f(x) = 8x^3 + 5x^2 + 7 = \Omega(x^3) \text{ [} k = 0 \text{ and } C = 8]$$

Definition 3: Let f and g be functions from the set of non-negative integers or the set of non-negative real numbers to the set of non-negative real numbers. We say that $f(x) = \theta(g(x))$ if $f(x) = O(g(x))$ and $f(x) = \Omega(g(x))$.

Example: Show that $3x^2 + 8x \log x = \theta(x^2)$.

Solution:

$$3x^2 + 8x \log x \leq 3x^2 + 8xx = 3x^2 + 8x^2 = 11x^2 \text{ [when } x \geq 1]$$

$$\therefore 3x^2 + 8x \log x = O(x^2) \text{ [} k = 1 \text{ and } C = 11]$$

$$3x^2 + 8x \log x \geq 3x^2 \text{ [when } x \geq 1]$$

$$\therefore 3x^2 + 8x \log x = \Omega(x^2) \text{ [} k = 1 \text{ and } C = 3]$$

$$\therefore 3x^2 + 8x \log x = \theta(x^2)$$

Theorem 4: Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, where $a_0, a_1, \dots, a_{n-1}, a_n$ are real numbers with $a_n \neq 0$. Then $f(x) = \theta(x^n)$.

Example:

$$3x^8 + 10x^7 + 221x^2 + 1444 = \theta(x^8)$$

$$x^{19} - 18x^4 - 10112 = \theta(x^{19})$$

Complexity of Algorithms

Time Complexity

The **time complexity** of an algorithm is an estimate of the time required by the algorithm to solve a problem of a particular size.

The time complexity of an algorithm can be expressed in terms of the number of operations used by the algorithm when the input has a particular size.

WORST-CASE COMPLEXITY

By the **worst-case** complexity of an algorithm, we mean the largest number of operations needed to solve the given problem using this algorithm on input of specified size.

AVERAGE-CASE COMPLEXITY

In **average-case** complexity, the average number of operations used to solve the problem over all inputs of a given size is found.

Example: Describe the time complexity of Algorithm 1 for finding the maximum element in a set.

Solution:

ALGORITHM 1: Finding the Maximum Element in a Finite Sequence.

procedure *max* ($a_1, a_2, \dots, a_n$: integers)

max := a_1

For $i := 2$ **to** n

if $max < a_i$ **then** $max := a_i$

{*max* is the largest element}

The **for** loop will be executed $(n - 1)$ times.

For each **for** loop, two comparisons will be done.

One more comparison will be needed to exit the **for** loop.

Therefore, exactly $2(n - 1) + 1 = 2n - 1$ comparisons are used.

Hence, the average-case time complexity of the algorithm is $2n - 1 = \theta(n)$.

Example: Describe the worst-case time complexity of the linear search algorithm.

Solution:

ALGORITHM 2: The Linear Search Algorithm.

```
procedure linear search ( $x$ : integer,  $a_1, a_2, \dots, a_n$ : distinct integers)
 $i := 1$ 
while ( $i \leq n$  and  $x \neq a_i$ )
     $i := i + 1$ 
if  $i \leq n$  then  $location := i$ 
else  $location := 0$ 
{ $location$  is the subscript of the term that equals  $x$ , or is 0 if  $x$  is not found}
```

At each step of the **while** loop, two comparisons are performed.
One more comparison is made outside the loop.

If $x = a_i$, then $2i + 1$ comparisons are used.

When the element is not in the list, $2n + 1 + 1 = 2n + 2$ comparisons are required.

Hence, the worst-case time complexity of linear search algorithm is $2n + 2 = O(n)$.

Example: Describe the time complexity of the binary search algorithm.

Solution:

ALGORITHM 3: The Binary Search Algorithm.

```
procedure binary search ( $x$ : integer,  $a_1, a_2, \dots, a_n$ : increasing integers)
 $i := 1$       { $i$  is left endpoint of search interval}
 $j := n$       { $j$  is right endpoint of search interval}
while  $i < j$ 
begin
     $m := \lfloor (i + j)/2 \rfloor$       {  $m$  is middle point of search interval}
    if  $x > a_m$  then  $i := m + 1$ 
    else  $j := m$ 
end
if  $x = a_i$  then  $location := i$ 
else  $location := 0$ 
{ $location$  is the subscript of the term that equals  $x$ , or is 0 if  $x$  is not found}
```

Example: Describe the time complexity of the binary search algorithm. (Contd.)

Assume $n = 2^k$, where k is a nonnegative integer

$$\therefore k = \log n$$

<u>Search Interval Length</u>	<u>Comparisons</u>
2^k	2
2^{k-1}	2
2^{k-2}	2
...	...
2^2	2
2^1	2
$2^0 = 1$	1

One more comparison is used to determine if this item is x .

Thus, binary search algorithm requires

$$2k + 1 + 1 = 2k + 2 = 2\log n + 2 \text{ comparisons.}$$

Hence, the average-case time complexity of binary search is $2\log n + 2 = \theta(\log n)$.

Example: Describe the average-case complexity of the linear search algorithm, assuming that the element x is in the list.

Solution:

ALGORITHM 2: The Linear Search Algorithm.

```
procedure linear search ( $x$ : integer,  $a_1, a_2, \dots, a_n$ : distinct integers)
 $i := 1$ 
while ( $i \leq n$  and  $x \neq a_i$ )
     $i := i + 1$ 
If  $i \leq n$  then  $location := i$ 
else  $location := 0$ 
{ $location$  is the subscript of the term that equals  $x$ , or is 0 if  $x$  is not found}
```

<u>Location of x</u>	<u>Comparison needed</u>
1	$2 \times 1 + 1 = 3$
2	$2 \times 2 + 1 = 5$
...	...
n	$2n + 1$

Example: Describe the average-case complexity of the linear search algorithm, assuming that the element x is in the list. (contd.)

Thus, the average number of comparisons used by the linear search algorithm (when x is known to be in the list) is

$$\frac{(2 \times 1 + 1) + (2 \times 2 + 1) + \dots + (2n + 1)}{n} = \frac{2(1 + 2 + \dots + n) + (1 + 1 + \dots + 1)}{n} = \frac{2\left[\frac{n(n+1)}{2}\right] + n}{n} = (n+1) + 1 = n + 2$$

Hence, the average-case time complexity of linear-search algorithm is $n + 2 = \theta(n)$.

The Integer and Division

Theorem 2: THE DIVISION ALGORITHM Let a be an integer and d a positive integer. Then there are unique integers q and r , with $0 \leq r < d$, such that $a = dq + r$

Definition 2: In the equality given in the division algorithm, d is called the *divisor*, a is called the *dividend*, q is called the *quotient*, and r is called the *remainder*. These notations are used to express the quotient and remainder:

$$q = a \text{ div } d, r = a \text{ mod } d.$$

Example: What are the quotient and remainder when 101 is divided by 11?

Solution:

$$101 = 11 \cdot 9 + 2$$

The quotient is $101 \text{ div } 11 = 9$

The remainder is $101 \text{ mod } 11 = 2$

Example: What are the quotient and remainder when -11 is divided by 3?

Solution:

$$-11 = 3(-4) + 1$$

The quotient is $-11 \text{ div } 3 = -4$

The remainder is $-11 \text{ mod } 3 = 1$

Modular Arithmetic

Definition 3: If a and b are integers and m is a positive integer, then a is *congruent to b modulo m* if m divides $a - b$. We use the notation $a \equiv b \pmod{m}$ to indicate that a is congruent to b modulo m . If a and b are not congruent modulo m , we write $a \not\equiv b \pmod{m}$.

Theorem 3: Let a and b be integers, and let m be a positive integer. Then $a \equiv b \pmod{m}$ if and only if $a \bmod m = b \bmod m$.

Example: Determine whether 17 is congruent to 5 modulo 6 and whether 24 and 14 are congruent modulo 6.

Solution:

6 divides $(17 - 5) = 12$. $\therefore 17 \equiv 5 \pmod{6}$

6 does not divide $(24 - 14) = 10$. $\therefore 24 \not\equiv 14 \pmod{6}$

Theorem 4: Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

Proof: Omitted.

The set of all integers congruent to an integer a modulo m is called the **congruence class** of a modulo m .

Theorem 5: Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

Proof: Omitted.

Example: Because $7 \equiv 2 \pmod{5}$ and $11 \equiv 1 \pmod{5}$, it follows from Theorem 5 that
 $7 + 11 = 18 \equiv 2 + 1 = 3 \pmod{5}$
 $7 \cdot 11 = 77 \equiv 2 \cdot 1 = 2 \pmod{5}$

Corollary 2: Let m be a positive integer and let a and b be integers. Then
 $(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$
and
 $ab \bmod m = ((a \bmod m)(b \bmod m)) \bmod m$.

Proof: Omitted.

Example:

$(8 + 14) \bmod 5 = 22 \bmod 5 = 2$
 $((8 \bmod 5) + (14 \bmod 5)) \bmod 5 = (3 + 4) \bmod 5 = 7 \bmod 5 = 2$

$(8 \cdot 14) \bmod 5 = 112 \bmod 5 = 2$
 $((8 \bmod 5)(14 \bmod 5)) \bmod 5 = (3 \cdot 4) \bmod 5 = 12 \bmod 5 = 2$

Application of Congruences

Example: Pseudorandom Numbers

The most commonly used procedure for generating pseudorandom numbers is the **linear congruential method**.

We choose four integers: the **modulus** m , **multiplier** a , **increment** c , and **seed** x_0 , with $2 \leq a < m$, $0 \leq c < m$, and $0 \leq x_0 < m$.

We generate a sequence of pseudorandom numbers $\{x_n\}$, with $0 \leq x_n < m$ for all n , by successively using the congruence

$$x_{n+1} = (ax_n + c) \bmod m.$$

Example: Pseudorandom Numbers (contd.)

Let $m = 9$, $a = 7$, $c = 4$, and $x_0 = 3$

$$x_{n+1} = (ax_n + c) \bmod m$$

$$x_0 = 3$$

$$x_1 = (7x_0 + 4) \bmod 9 = (7 \cdot 3 + 4) \bmod 9 = 25 \bmod 9 = 7$$

$$x_2 = (7x_1 + 4) \bmod 9 = (7 \cdot 7 + 4) \bmod 9 = 53 \bmod 9 = 8$$

$$x_3 = (7x_2 + 4) \bmod 9 = (7 \cdot 8 + 4) \bmod 9 = 60 \bmod 9 = 6$$

$$x_4 = (7x_3 + 4) \bmod 9 = (7 \cdot 6 + 4) \bmod 9 = 46 \bmod 9 = 1$$

$$x_5 = (7x_4 + 4) \bmod 9 = (7 \cdot 1 + 4) \bmod 9 = 11 \bmod 9 = 2$$

$$x_6 = (7x_5 + 4) \bmod 9 = (7 \cdot 2 + 4) \bmod 9 = 18 \bmod 9 = 0$$

$$x_7 = (7x_6 + 4) \bmod 9 = (7 \cdot 0 + 4) \bmod 9 = 4 \bmod 9 = 4$$

$$x_8 = (7x_7 + 4) \bmod 9 = (7 \cdot 4 + 4) \bmod 9 = 32 \bmod 9 = 5$$

$$x_9 = (7x_8 + 4) \bmod 9 = (7 \cdot 5 + 4) \bmod 9 = 39 \bmod 9 = 3$$

$$x_{10} = (7x_9 + 4) \bmod 9 = (7 \cdot 3 + 4) \bmod 9 = 25 \bmod 9 = 7$$

Primes

Definition 1: A positive integer p greater than 1 is called a *prime* if the only positive factors of p are 1 and p . A positive integer that is greater than 1 and is not prime is called *composite*.

Example: The integer 7 is prime since its only positive factors are 1 and 7, whereas the integer 9 is composite since it is divisible by 3.

Theorem 1: THE FUNDAMENTAL THEOREM OF ARITHMETIC Every positive integer greater than 1 can be written uniquely as a prime or as the product of two or more primes where the prime factors are written in order of non-decreasing size.

Proof: Omitted.

Example: The prime factorizations of 100, 641, 999, and 1024 are given by

$$100 = 2 \cdot 2 \cdot 5 \cdot 5 = 2^2 \cdot 5^2$$

$$641 = 641$$

$$999 = 3 \cdot 3 \cdot 3 \cdot 37 = 3^3 \cdot 37^1$$

$$1024 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^{10}$$

Theorem 2: If n is a composite integer, then n has at least one prime divisor less than or equal to $\sqrt{n}$.

Proof. Omitted.

Example: Show that 101 is prime.

Solution:

The only primes not exceeding $\sqrt{101} = 10$ are 2, 3, 5, and 7. Since 101 is not divisible by 2, 3, 5, or 7, it follows that 101 is prime.

Example: Find the prime factorization of 7007.

Solution:

First, perform divisions of 7007 by successive primes, beginning with 2 but not exceeding $\sqrt{7007} = 83$:
 $7007 / 7 = 1001$

Next, divide 1001 by successive primes, beginning with 7 but not exceeding $\sqrt{1001} = 31$:
 $1001 / 7 = 143$

Next, divide 143 by successive primes, beginning with 7 but not exceeding $\sqrt{143} = 11$:
 $143 / 11 = 13$

Next, divide 13 by successive primes, beginning with 11 but not exceeding $\sqrt{13} = 3$, which is not possible. The procedure is completed.

Therefore, $7007 = 7 \cdot 7 \cdot 11 \cdot 13 = 7^2 \cdot 11^1 \cdot 13^1$

Greatest Common Divisors and Least Common Multiples

Definition 2: Let a and b be integers, not both zero. The largest integer d such that $d \mid a$ and $d \mid b$ is called the *greatest common divisor* of a and b . The greatest common divisor of a and b is denoted by $\gcd(a, b)$.

Example: What is the greatest common divisor of 24 and 36?

Solution:

The positive common divisors of 24 and 36 are 1, 2, 3, 4, 6, and 12. Hence, $\gcd(24, 36) = 12$.

Example: What is the greatest common divisor of 17 and 22?

Solution:

The only common divisor of 17 and 22 is 1, hence $\gcd(17, 22) = 1$.

Definition 3: The integers a and b are *relatively prime* if their greatest common divisor is 1.

Example: The integers 17 and 22 are relatively prime, since $\gcd(17, 22) = 1$.

Definition 4: The integers $a_1, a_2, \dots, a_n$ are *pair wise relatively prime* if $\gcd(a_i, a_j) = 1$ whenever $1 \leq i < j \leq n$.

Example: Determine whether the integers 10, 17, and 21 are pairwise relatively prime and whether the integers 10, 19, and 24 are pairwise relatively prime.

Solution:

$\gcd(10, 17) = 1$, $\gcd(10, 21) = 1$, and $\gcd(17, 21) = 1$. Therefore, 10, 17, and 21 are pairwise relatively prime.

$\gcd(10, 24) = 2 > 1$. Therefore, 10, 19, and 24 are not pairwise relatively prime.

One way to find the greatest common divisor of two integers is to use the prime factorizations of these integers.

Suppose that the prime factorizations of the integers a and b , neither equal to zero, are

$$a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}$$

$$b = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n}$$

where exponent is a nonnegative integer, and where all primes occurring in the prime factorization of either a or b are included in both factorization, with zero exponents if necessary. Then

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \cdots p_n^{\min(a_n, b_n)}$$

Example: The prime factorization of 120 and 500 are

$$120 = 2^3 \cdot 3^1 \cdot 5^1$$

$$500 = 2^2 \cdot 3^0 \cdot 5^3$$

$$\gcd(120, 500) = 2^{\min(3, 2)} 3^{\min(1, 0)} 5^{\min(1, 3)} = 2^2 3^0 5^1 = 20$$

Definition 5: The least common multiple of the positive integers a and b is the smallest positive integer that is divisible by both a and b . The least common multiple of a and b is denoted by $\text{lcm}(a, b)$.

Suppose that the prime factorizations of the integers a and b , neither equal to zero, are

$$a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}$$

$$b = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n}$$

where exponent is a nonnegative integer, and where all primes occurring in the prime factorization of either a or b are included in both factorization, with zero exponents if necessary. Then

$$\text{lcm}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \cdots p_n^{\max(a_n, b_n)}$$

Example: What is the least common multiple of $95256 = 2^3 \cdot 3^5 \cdot 7^2$ and $432 = 2^4 \cdot 3^3$?

Solution:

$$\text{lcm}(2^3 \cdot 3^5 \cdot 7^2, 2^4 \cdot 3^3) = \text{lcm}(2^3 \cdot 3^5 \cdot 7^2, 2^4 \cdot 3^3 \cdot 7^0) = 2^{\max(3, 4)} 3^{\max(5, 3)} 7^{\max(2, 0)} = 2^4 \cdot 3^5 \cdot 7^2 = 190512$$

Graphs

Graphs and Graph Models

Definition 1: A *graph* $G = (V, E)$ consists of V , a nonempty set of *vertices* (or *nodes*) and E , a set of *edges*. Each edge has either one or two vertices associated with it, called its *endpoints*. An edge is said to *connect* its end points.

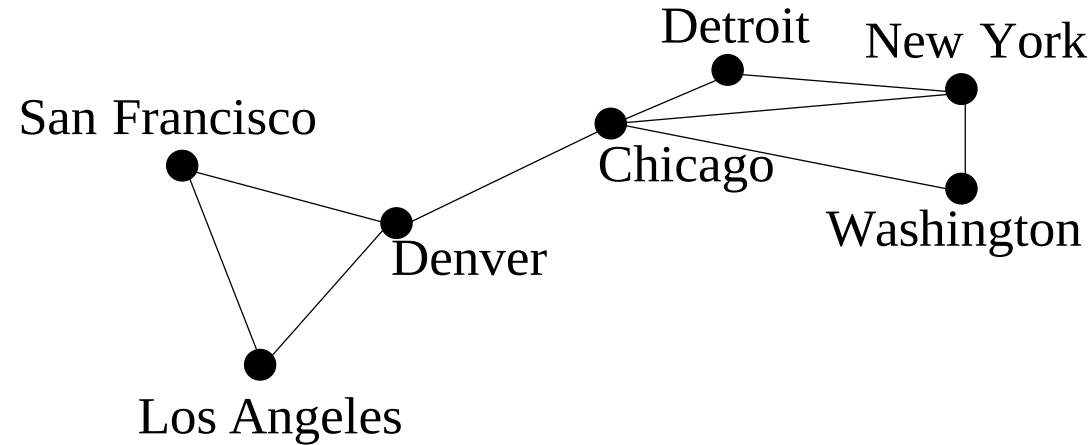


Figure: A computer network represented by a graph.

When there is an edge of a graph associated to $\{u, v\}$, we can say that $\{u, v\}$ is an edge of the graph.

If an edge connects a vertex to itself, it is called a **loop**.

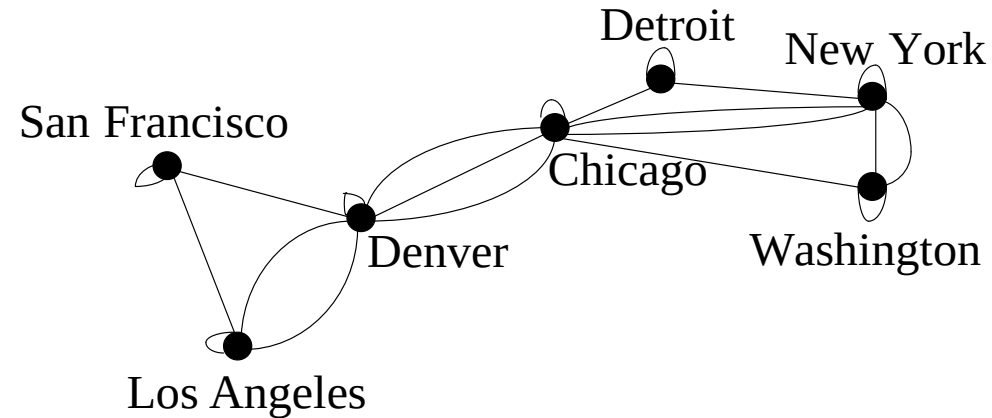


Figure: A graph representing a computer network with multiple edges and loops.

So far the graphs we have introduced are **undirected graphs**. Their edges are also said to be **undirected**.

Definition 2: A *directed graph* (or *digraph*) $G = (V, E)$ consists of a nonempty set of vertices V and a set of *directed edges* (or *arcs*) E . Each directed edge is associated with an ordered pair of vertices. The directed edge associated with the ordered pair (u, v) is said to *start* at u and *end* at v .

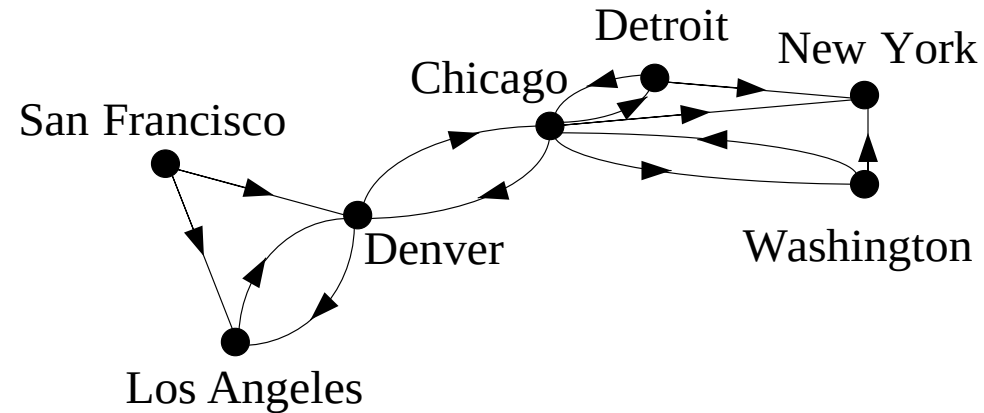


Figure: A graph representing a communication network with one-way communications links.

We use an arrow pointing from u to v to indicate the direction of the edge that starts at u and ends at v .

We call (u, v) an edge if there is an edge associated to it in the graph.

A graph with both directed and undirected edges is called a **mixed graph**.

Graph Terminology and Special Types of Graphs

Definition 1: Two vertices u and v in an undirected graph G are called *adjacent* (or *neighbors*) in G if u and v are endpoints of an edge of G . If e is associated with $\{u, v\}$, the edge e is called *incident with* the vertices u and v . The edge e is also said to *connect* u and v . The vertices u and v are called *endpoints* of an edge associated with $\{u, v\}$.

Definition 2: The *degree of a vertex in an undirected graph* is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex. The degree of the vertex v is denoted by $\deg(v)$.

Example: What are the degrees of the vertices in the graph G and H displayed in Figure 1?

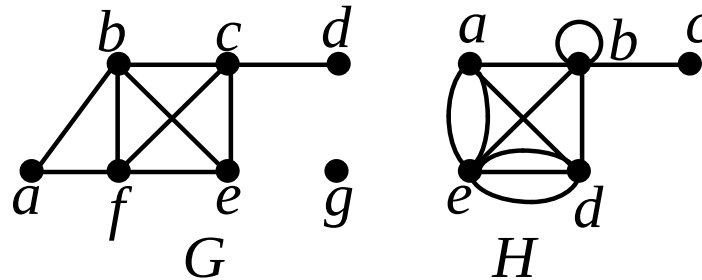


Figure: The Undirected Graphs G and H .

Solution:

In G , $\deg(a) = 2$, $\deg(b) = 4$, $\deg(c) = 4$, $\deg(d) = 1$, $\deg(e) = 3$, $\deg(f) = 4$, $\deg(g) = 0$

In H , $\deg(a) = 4$, $\deg(b) = 6$, $\deg(c) = 1$, $\deg(d) = 5$, $\deg(e) = 6$

A vertex of degree zero is called **isolated**. A vertex of degree one is called **pendant**.

Theorem 1: THE HANDSHAKING THEOREM

Let $G = (V, E)$ be an undirected graph with $|E|$ edges. Then

$$2|E| = \sum_{v \in V} \deg(v)$$

Example: How many edges are there in a graph with 10 vertices each of degree six?

Solution:

Sum of degrees of the vertices = $6 \cdot 10 = 60$

$$2|E| = 60$$

$$\therefore |E| = \frac{60}{2} = 30$$

Definition 3: When (u, v) is an edge of the graph G with directed edges, u is said to be *adjacent to* v and v is said to be *adjacent from* u . The vertex u is called the *initial* or *start vertex* of (u, v) , and v is called the *terminal* or *end vertex* of (u, v) . The initial vertex and terminal vertex of a loop are the same.

Definition 4: In a graph with directed edges the *in-degree* of a vertex v , denoted by $\deg^-(v)$, is the number of edges with v as their terminal vertex. The *out-degree* of v , denoted by $\deg^+(v)$, is the number of edges with v as their initial vertex. (Note that a loop at a vertex contributes 1 to both the in-degree and the out-degree of this vertex.)

Example: Find the in-degree and out-degree of each vertex in the graph G with directed edges shown in Figure 2.

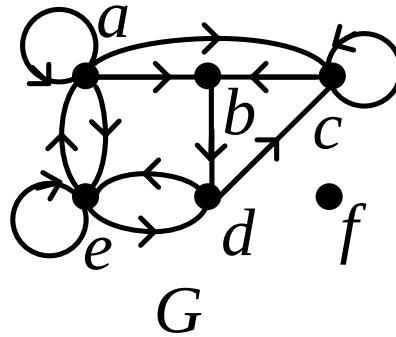


Figure: The Directed Graph G .

Solution:

The in-degree in G are: $\deg^-(a) = 2$, $\deg^-(b) = 2$, $\deg^-(c) = 3$, $\deg^-(d) = 2$, $\deg^-(e) = 3$, $\deg^-(f) = 0$

The out-degree in G are: $\deg^+(a) = 4$, $\deg^+(b) = 1$, $\deg^+(c) = 2$, $\deg^+(d) = 2$, $\deg^+(e) = 3$, $\deg^+(f) = 0$

Theorem 3: Let $G = (V, E)$ be a graph with directed edges. Then

$$\sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v) = |E|$$

Some Special Graphs

Example: Complete Graphs

The **complete graph on n vertices**, $n \geq 1$, denoted by K_n , is the graph that contains exactly one edge between each pair of distinct vertices.

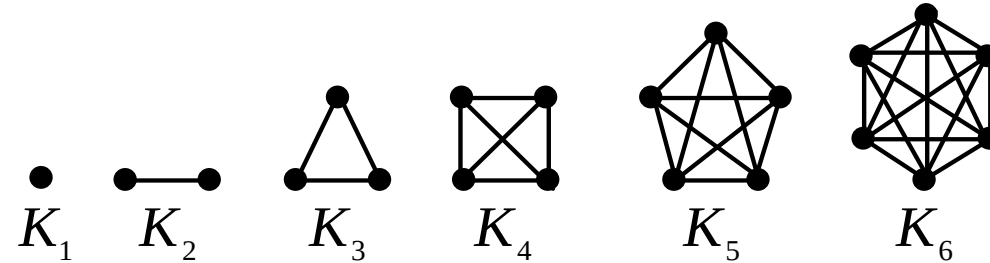


Figure: The Graphs K_n for $1 \leq n \leq 6$.

Example: Cycles

The **cycle C_n** , $n \geq 3$, consists of n vertices $v_1, v_2, \dots, v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}$, and $\{v_n, v_1\}$.

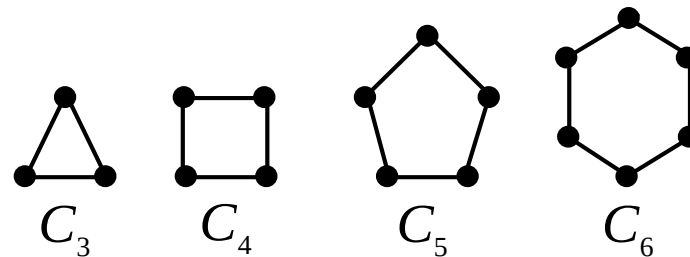


Figure: The Cycles C_3, C_4, C_5 , and C_6 .

Example: Wheels

We obtain the **wheel** W_n when we add an additional vertex to the cycle C_n , for $n \geq 3$, and connect this new vertex to each of the n vertices in C_n , by new edges.

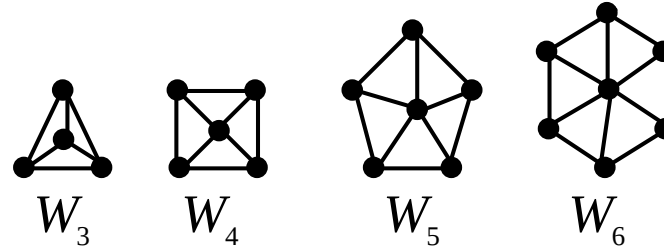


Figure: The Wheels W_3 , W_4 , W_5 , and W_6 .

Bipartite Graphs

Definition 5: A simple graph G is called *bipartite* if its vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 and a vertex in V_2 (so that no edge in G connects either two vertices in V_1 or two vertices in V_2). When this condition holds, we call the pair (V_1, V_2) a *bipartition* of the vertex set V of G .

Example: C_6 is bipartite as shown in Figure 7.

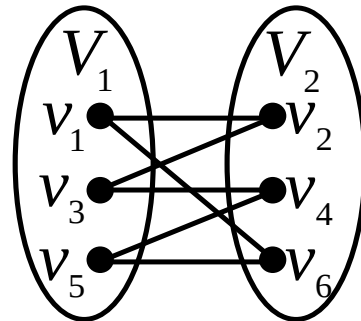
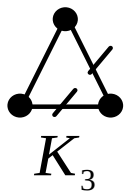


Figure: Showing that C_6 is Bipartite.

Example: K_3 is not bipartite.



Example. Are the graphs G and H displayed in Figure 8 bipartite?

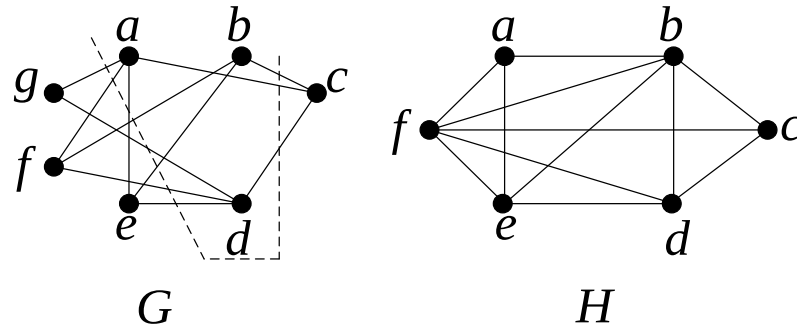


Figure.: The Undirected Graphs G and H .

Solution:

Graph G is bipartite, since its vertex set is the union of two disjoint subsets, $\{a, b, d\}$ and $\{c, e, f, g\}$, and each edge connects a vertex in one of these subsets to a vertex in the other subset.

Graph H is not bipartite, since its vertex set cannot be partitioned into two subsets so that edges do not connect two vertices from the same subset.

Representing Graphs and Graph Isomorphism

Representing Graphs

Adjacency Lists

One way to represent a graph is to use **adjacency lists**, which specify the vertices that are adjacent to each vertex of the graph.

Example: Use adjacency lists to describe the graph given in Figure 1.

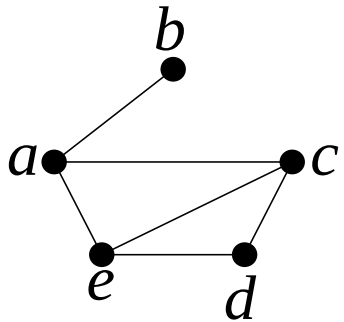


Figure:

Solution:

Table: An Edge List for a Graph.

Vertex	Adjacent vertices
<i>a</i>	<i>b, c, e</i>
<i>b</i>	<i>a</i>
<i>c</i>	<i>a, d, e</i>
<i>d</i>	<i>c, e</i>
<i>e</i>	<i>a, c, d</i>

Example: Represent the directed graph shown in Figure 2 by listing all the vertices that are the terminal vertices of edges starting at each vertex of the graph.

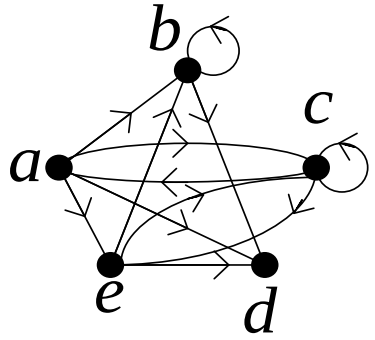


Figure:. A Directed Graph.

Solution:

Table 2. An Edge List for a Directed Graph.

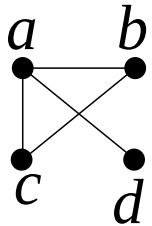
Initial vertex	Terminal Veritces
<i>a</i>	<i>b, c, d, e</i>
<i>b</i>	<i>b, d</i>
<i>c</i>	<i>a, c, e</i>
<i>d</i>	
<i>e</i>	<i>b, c, d</i>

Adjacency Matrices

Suppose that $G = (V, E)$ is a graph where $|V| = n$. Suppose that the vertices of G are listed arbitrarily as $v_1, v_2, \dots, v_n$. The **adjacency matrix** $\mathbf{A}$ of G , with respect to this listing of the vertices, is an $n \times n$ zero-one matrix with 1 as its (i, j) th entry when v_i and v_j are adjacent, and 0 as its (i, j) th entry when they are not adjacent. In other words, if its adjacency matrix is $\mathbf{A} = [a_{ij}]$, then

$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \text{ is an edge of } G \\ 0 & \text{otherwise} \end{cases}$$

Example: Use an adjacency matrix to represent the graphs shown in Figure 3.



Solution:

We order the vertices as a, b, c, d . The matrix representing this graph is

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Figure: A Simple Graph.

Example: Draw a graph with the adjacency matrix

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

with respect to the ordering of vertices a, b, c, d .

Solution:

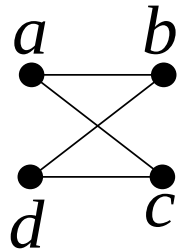
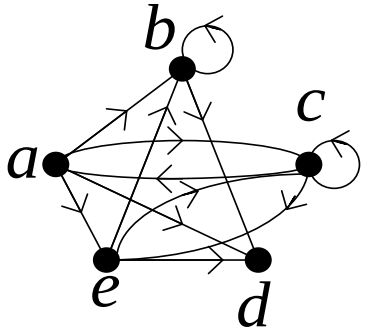


Figure:. A Graph with the Given Adjacency Matrix.

The matrix for a directed graph $G = (V, E)$ is $\mathbf{A} = [a_{ij}]$, where

$$a_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \text{ is an edge of } G \\ 0 & \text{otherwise} \end{cases}$$

Example: Represent the directed graph shown in Figure 2 by adjacency matrix.



Solution: We order the vertices as a, b, c, d, e . The matrix representing this graph is

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

Figure: A Directed Graph.

Trees

Introduction to Trees

Definition 1: A *tree* is a connected undirected graph with no simple circuits.

Example: Which of the graphs shown in Figure 2 are trees?

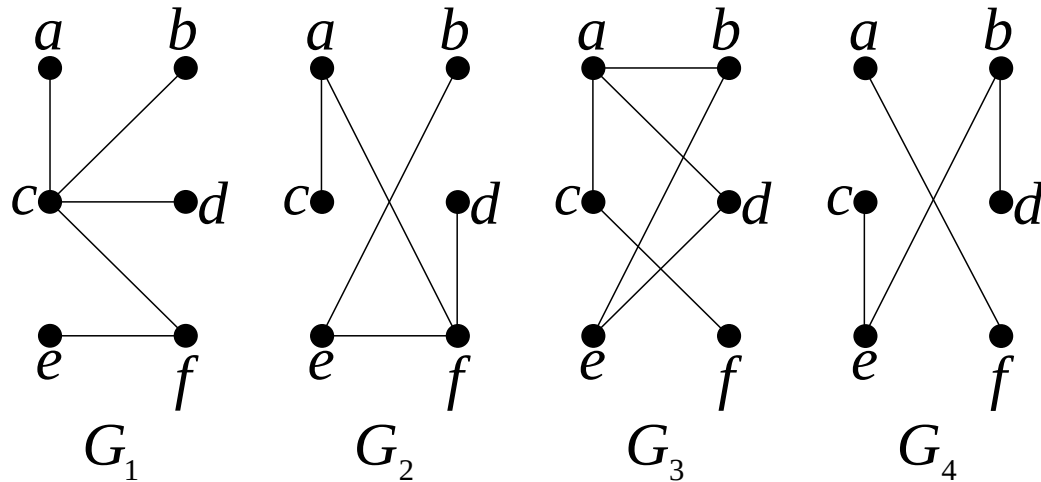


Figure: Examples of Trees and Graphs That Are Not Trees.

Solution:

G_1 and G_2 are trees, since both are connected graphs with no simple circuits.

G_3 is not a tree because e, b, a, d, e is a simple circuit in this graph.

G_4 is not a tree since it is not connected.

Theorem 1: An undirected graph is a tree if and only if there is a unique simple path between any two of its vertices.

Definition 2: A *rooted tree* is a tree in which one vertex has been designated as the root and every edge is directed away from the root.

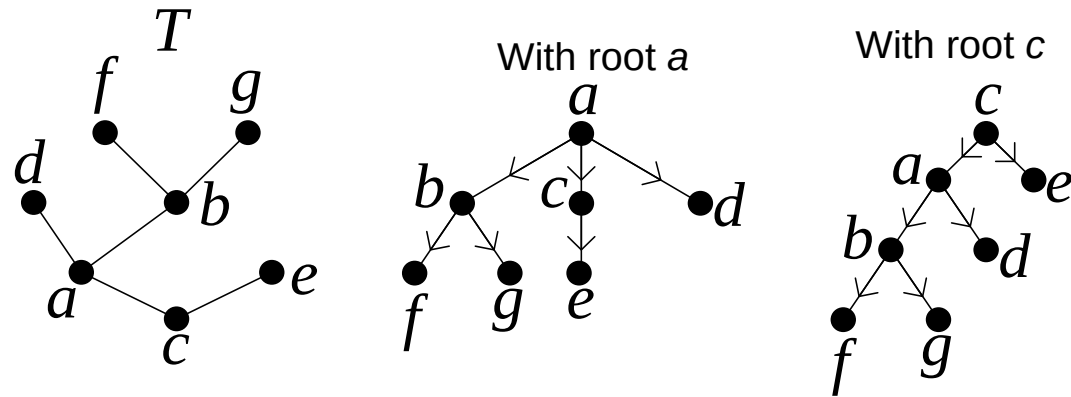


Figure: A tree and Rooted Trees Formed by Designating Two Roots.

We usually draw a rooted tree with its root at the top of the graph. The arrows indicating the directions of the edges in a rooted tree can be omitted, since the choice of root determines the direction of the edges.

Suppose that T is a rooted tree.

If v is a vertex in T other than the root, the **parent** of v is the unique vertex u such that there is a directed edge from u to v .

When u is the parent of v , v is called a **child** of u .

Vertices with the same parent are called **siblings**.

The **ancestors** of a vertex other than the root are the vertices in the path from the root to this vertex, excluding the vertex itself and including the root.

The **descendants** of a vertex v are those vertices that have v as an ancestor.

A vertex of a tree is called a **leaf** if it has no children.

Vertices that have children are called **internal vertices**.

If a is a vertex in a tree, the **subtree** with a as its root is the subgraph of the tree consisting of a and its descendants and all edges incident to these descendants.

Example: In the rooted tree T (with root a) shown in Figure 5, find the parent of c , the children of g , the siblings of h , all ancestors of e , all descendants of b , all internal vertices, and all leaves. What is the subtree rooted at g ?

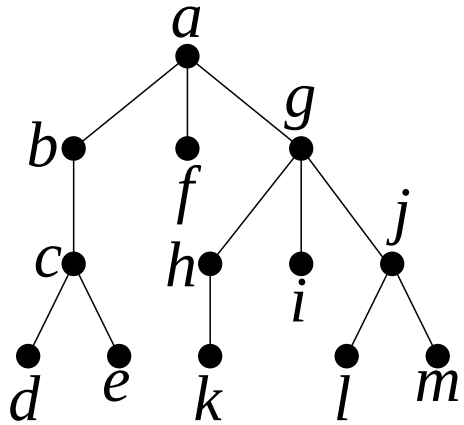


Figure: A Rooted Tree H .

The subtree rooted at g is shown in Figure 6.

Solution:

The parent of c is b .

The children of g are h , i , and j .

The siblings of h are i and j .

The ancestors of e are c , b , and a .

The descendants of b are c , d , and e .

The internal vertices are a , b , c , g , h , and j .

The leaves are d , e , f , i , k , l , and m .

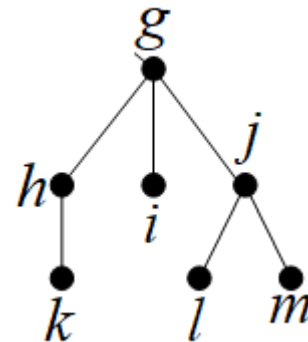


Figure 6. The Subtree Rooted at g .

Definition 3: A rooted tree is called an m -ary tree if every internal vertex has no more than m children. The tree is called a *full m -ary tree* if every internal vertex has exactly m children. An m -ary tree with $m = 2$ is called a *binary tree*.

Example:

Are the rooted trees in Figure 7 full m -ary trees for some positive integer m ?

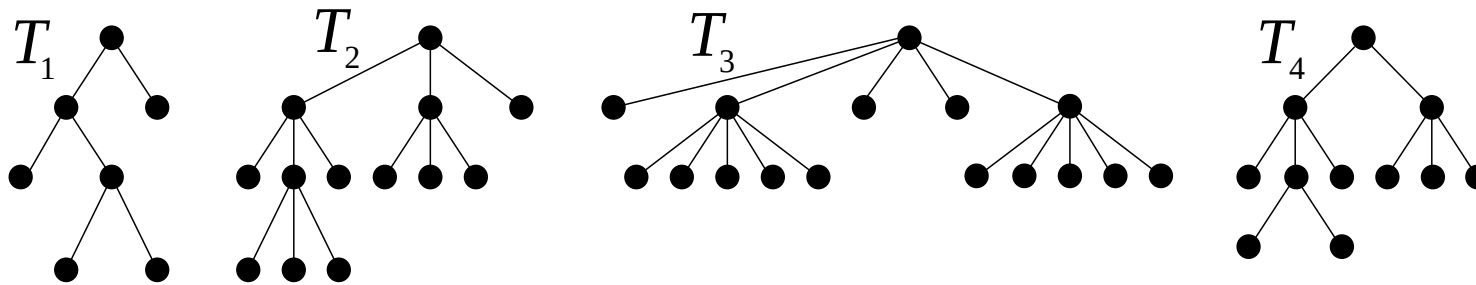


Figure: Four Rooted Trees.

Solution:

T_1 is a full binary tree.

T_2 is a full 3-ary tree.

T_3 is full 5-ary tree.

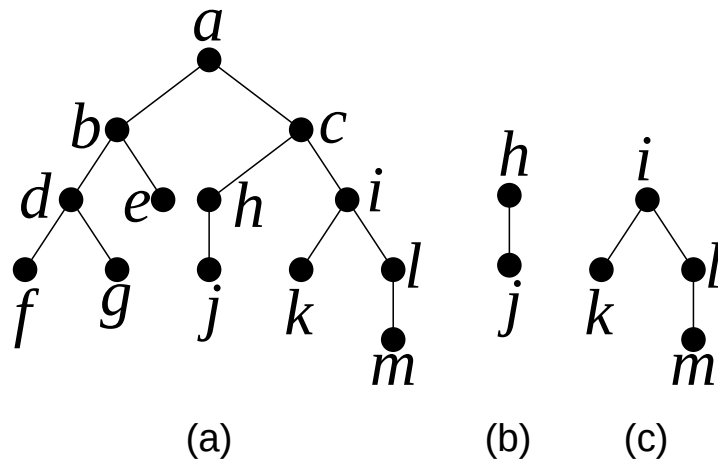
T_4 is not a full m -ary tree for any m since some of its internal vertices have two children and others have three children.

An **ordered rooted tree** is a rooted tree where the children of each internal vertex are ordered. Ordered rooted trees are drawn so that the children of each internal vertex are shown in ordered from left to right.

In an ordered binary tree (usually called just a **binary tree**), if an internal vertex has two children, the first child is called the **left child** and the second child is called the **right child**.

The tree rooted at the left child of a vertex is called the **left subtree** of this vertex, and the tree rooted at the right child of a vertex is called the **right subtree** of the vertex.

Example: What are the left and right children of d in the binary tree T shown in Figure 8(a)? What are the left and right subtrees of c ?



Solution:

The left child of d is f and the right child is g .

We show the left and right subtrees of c in Figures 8(b) and 8(c), respectively.

Figure: A Binary Tree T and Left and Right Subtrees of Vertex c .

Properties of Trees

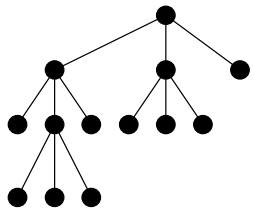
Theorem 2: A tree with n vertices has $n - 1$ edges.

Theorem 3: A full m -ary tree with i internal vertices contains $n = mi + 1$ vertices.

Theorem 4: A full m -ary tree with

- (i) n vertices has $i = (n - 1)/m$ internal vertices and $l = [(m - 1)n + 1]/m$ leaves,
- (ii) i internal vertices has $n = mi + 1$ vertices and $l = (m - 1)i + 1$ leaves,
- (iii) l leaves has $n = (ml - 1)/(m - 1)$ vertices and $i = (l - 1)/(m - 1)$ internal vertices.

Example:



The above tree is a full 3-ary tree, that is, $m = 3$.

The number of vertices is $n = 13$.

The number of internal vertices is $i = 4$.

The number of leaves is $l = 9$.

From Theorem 2,

The number of edges is $e = (n - 1) = 13 - 1 = 12$.

From Theorem 4,

(i) $i = (n - 1)/m = (13 - 1)/3 = 4$

$$l = [(m - 1)n + 1]/m = [(3 - 1)13 + 1]/3 = 9$$

(ii) $n = mi + 1 = 3 \times 4 + 1 = 13$

$$l = (m - 1)i + 1 = (3 - 1)4 + 1 = 9$$

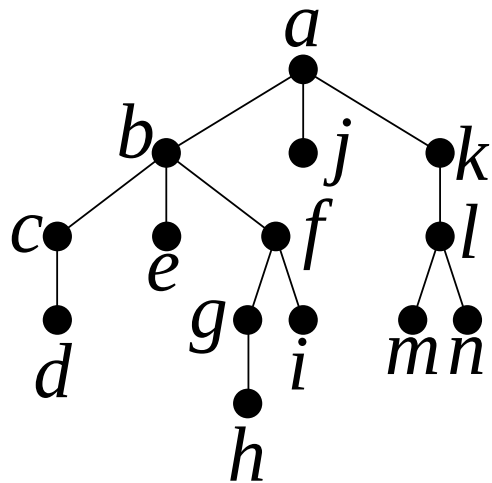
(iii) $n = (ml - 1)/(m - 1) = (3 \times 9 - 1)/(3 - 1) = 13$

$$i = (l - 1)/(m - 1) = (9 - 1)/(3 - 1) = 4$$

The **level** of a vertex v in a rooted tree is the length of the unique path from the root to this vertex. The level of the root is defined to be zero.

The **height** of a rooted tree is the maximum of the levels of vertices. In other words, the height of a rooted tree is the length of the longest path from the root to any vertex.

Example: Find the level of each vertex in the rooted tree shown in Figure 13. What is the height of this tree?



Solution:

The root a is at level 0.

Vertices b , j , and k are at level 1.

Vertices c , e , f , and l are at level 2.

Vertices d , g , i , m , and n are at level 3.

Finally, vertex h is at level 4.

Since the largest level of any vertex is 4, this tree has height 4.

Figure: A Rooted Tree.

A rooted m -ary tree of height h is **balanced** if all leaves are at levels h or $h - 1$.

Example: Which of the rooted trees shown in Figure 14 are balanced?

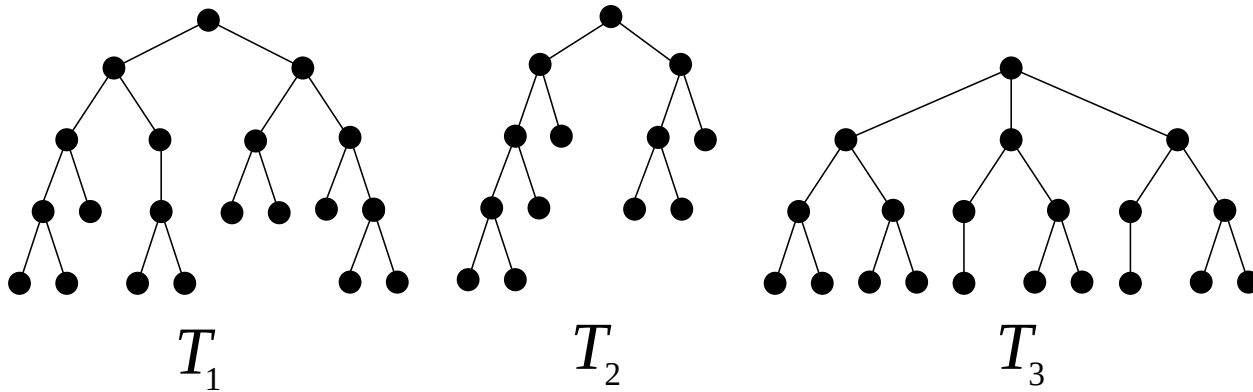


Figure: Some Rooted Trees.

Solution:

T_1 is balanced, since all its leaves are at levels 3 and 4.

T_2 is not balanced, since it has leaves at levels 2, 3, and 4.

T_3 is balanced, since all its leaves are at level 3.